

Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Motivation

- **Economic Problem:** Selecting best options under costly evaluation
 - Determining each option's true quality costs decision-maker time and effort
 - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
 - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
 - Graduate programs invest time and money into students' development
 - If programs are undervaluing applicants with certain characteristics, future such applicants may receive formerly-withheld opportunities

Research Questions

- ① Are admissions committees optimizing, or are they making mistakes?
 - Goal is to accept applicants with highest success probabilities who will matriculate
 - Per literature, measure success with program completion and job placement ranking
 - Predictors of acceptance only (success only) may be overweighted (underweighted)
 - ② If they are making mistakes, how can they make better admissions decisions?
 - In turn, by changing decision rules, how much better of a cohort can they admit?
- ★ This paper's methods generalize to admissions, hiring, and other settings

Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
 - Statistical tests of whether committee's decision rule aligns with desired objective
 - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
 - For many applicants, estimated success probabilities align with those implied by committee's decisions. But committee misweights some applicant characteristics
 - Correctly weighting these characteristics would significantly increase admitted cohort's average success probability without reading more application files

Literature Review

- **Graduate Admissions:** Regress acceptance and success on applicant characteristics
 - **Original Literature:** Attiyeh & Attiyeh (1997); Krueger & Wu (2000); Grove & Wu (2007); Athey, Katz, Krueger, Levitt & Poterba (2007)
 - **Study of 2002 Cohort:** Stock, Finegan & Siegfried (2006–2015)
 - **New Predictors/Methods:** Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
 - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
 - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)
- ★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

Synthetic Data Protocol (IRB-Approved)

① Administrative office receives data from university departments

- Algorithmically extracts variables from textual files, such as recommendation letters

② Synthetic data (Patki et al. 2016) used for model development

- Estimate joint distribution of variables, and simulate synthetic dataset from it
- Rows do not correspond to real subjects, preventing identity disclosure
- Cells with < 5 comparable observations binned to prevent attribute disclosure

③ Administrative office executes final estimation code on actual data

- Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

Outline

- 1 Regression Analysis
- 2 Structural Models
- 3 Structural Estimation
- 4 Optimality Test

Data Description

- **Sample:** Applicants to R1 university's economics PhD program from 2015–2019
- **Dependent Variables:** Measures of admission and success
 - **Admission:** Indicators for all stages of admissions process, including first-round filter
 - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Independent Variables:** Objective vs. subjective
 - **Objective:** Demographics and performance measures, such as GRE scores
 - **Subjective:** Textual processing of recommendation letters, application essays, CVs, and supplemental forms using dictionaries to construct variables

Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
 - **Success:** *Top 100 Placement* = 1 for 20% of applicants vs. 34% of matriculants
- **Finding:** Different independent variables best predict each dependent variable
 - **Accept:** *Committee Score* is best predictor. Other predictors include demographics, *GRE Quantitative Percentile*, *Work Experience*, and *CV Coding* (negative)
 - **Matriculate:** Predictors of acceptance often negatively predict matriculation. However, best predictor of matriculation is *Open House* attendance
 - **Success:** *PhD Program Rank* is best predictor. Other predictors include *Age*, *TOEFL/IELTS*, *Work Experience*, and *Essay Positive Terms (%)*

Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	T100 Placement
Winsorized Age	-0.0001	-0.0223**	0.0053*
Female	0.0163**	-0.0490	-0.0032
U.S.	-0.0013	0.1472*	0.0192
International Asian	-0.0214***	0.1644**	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0409*
Work Experience	0.0140**	0.1591**	0.0358**
Committee Score	0.0788***	-0.0371	0.0007
Advanced Math	0.0246	-0.1950**	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0159
CV Coding	-0.0044*	-0.0414**	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0123*
Open House		0.2615***	
Program Rank			-0.0011***
Missing Program Rank			-0.3965***

Acceptance vs. Success AMEs

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
#Professor Recommenders	-0.0033	-0.0069
Missing TOEFL/IELTS	0.0070	-0.0008
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality

» Regressions vs. Model

Insufficiency of Comparing AMEs

- Cannot test optimality by comparing AMEs across regressions. Suppose:
 - Work experience has large effect on success, graduate degree smaller effect
 - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
 - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
 - **Acceptance Regression:** $AME_{\text{Experience}} = 1, AME_{\text{Degree}} = 0$
 - **Success Regression:** $1 > AME_{\text{Experience}} > AME_{\text{Degree}} > 0$
 - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

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Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
 - Per literature, committee should accept applicants with highest success probabilities
 - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
 - Quantify gains from deviating from actual to counterfactual decision rule

Model Framework

- **Portfolio Choice Problem:** Accept subset of applicants with highest sum of success probabilities subject to capacity constraint on number of acceptances
- For each applicant $n \in \{1, \dots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^N \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_A,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success}, \text{Failure}\}$,
 N = number of applicants, and N_A = maximum number of acceptances

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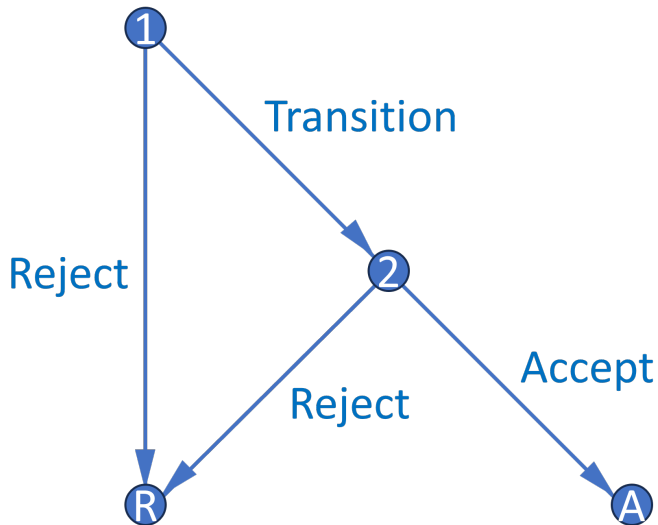
- BEMS (2022) prove that portfolio choice problem reduces to cutoff rule problem
 - Accept applicant if and only if their success probability exceeds cutoff probability
 - Decisions across applicants now independent, so no need to consider all subsets

►► Portfolio Choice to Cutoff Rule

Cutoff Rule Problem

- Start with two-round cutoff rule problem that accounts for costly evaluation:
 - **Round 1:** See only objective variables (e.g. GRE). Can transition application file to Round 2 to see subjective variables (e.g. recommendation letters), or can reject
 - **Round 2:** If chose to transition application file in Round 1, can accept or reject
 - Assume for now committee transitions and accepts optimal number of files
- Along with the baseline dynamic model, there are additional models
 - Static model is simpler and has interpretable cutoff probabilities in both rounds
 - Matriculation models account for both success and matriculation probabilities

Decision Graph



Dynamic Model

- **States:** r = round, x = objective variables, z = subjective variables, t = year, $\bar{k}$ = fixed program rank, ϵ_r = T1EV preference shock with scale parameter σ
- **Controls:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, y_r^* = marginal outcome

Dynamic Model

- **States:** $r = \text{round}$, $x = \text{objective variables}$, $z = \text{subjective variables}$, $t = \text{year}$, $\bar{k} = \text{fixed program rank}$, $\epsilon_r = \text{T1EV preference shock with scale parameter } \sigma$
- **Controls:** $d_1 \in \{\text{Transition, Reject}\}$, $d_2 \in \{\text{Accept, Reject}\}$
- **Outcomes:** $y \in \{\text{Success, Failure}\}$, $y_r^* = \text{marginal outcome}$

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Dynamic Model

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(e^{P(y=S|x, \tilde{z}, t, \bar{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma} \right) dF(\tilde{z}|x, t)$$

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Conditional Choice Probabilities:

$$P(d_r|x, (z), t, \bar{k}) = \frac{\exp(u_r(x, (z), t, \bar{k}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(x, (z), t, \bar{k}, \tilde{d}_r)/\sigma)}$$

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$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

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Static Model

- **Motivation:** Simpler model with interpretable cutoff probabilities in both rounds

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- $P(y = S | x, t, \bar{k})$ estimates inconsistent, but fitted values account for z 's effect on y

Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants
- For each applicant $n \in \{1, \dots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N \underbrace{P(y_n = S | x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^N \underbrace{P(m_n = M | x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where (x, z) = variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$,
 N = number of applicants, and N_M = maximum number of matriculants

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Theorem 1 (Proof)

*For large N , this problem has a cutoff rule solution on **success** probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- However, matriculation probabilities matter in Round 1 of a two-round model

Matriculation-Dynamic Model

Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

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$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(e^{[P(y=S|x, \tilde{z}, t, \bar{k}) - P(y_2^*=S|t)]/\sigma} + 1 \right) P(m=M|x, \tilde{z}, t) dF(\tilde{z} | x, t)$$

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Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\ \prod_{n=1}^N \underbrace{[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})]^{\mathbb{1}(y_n=S)} [(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- Do success probabilities implied by outcomes align with those implied by CCPs?

Cutoff Probability Identification

Theorem 2 (Proof)

The maximum likelihood cutoff probability estimate is such that the number of acceptances N_A equals the expected N_A . Also, for any cutoff probability and N_A , the likelihood is highest when the committee accepts the best N_A applicants.

- Identify each year's cutoffs by how selective committee is that year
- While selectivity identifies cutoffs, sorting improves restricted likelihood
 - If committee engages in sorting, Akaike Information Criterion more favorable
 - $AIC = 2[K - \log(\hat{\mathcal{L}}^R)]$, such that $K = \dim(\theta^R)$

Restricted Static Estimates

	R1 Cutoff Probability	R2 Cutoff Probability	Sigma
2015	29.36% (23.20%, 36.38%)	39.20% (32.75%, 46.05%)	
2016	29.17% (22.21%, 37.27%)	40.53% (33.55%, 47.92%)	
2017	37.06% (32.10%, 42.29%)	41.69% (34.83%, 48.90%)	
2018	38.12% (33.98%, 42.44%)	39.30% (31.80%, 47.33%)	
2019	34.04% (28.67%, 39.85%)	39.62% (32.54%, 47.16%)	
Sigma			0.0064 (0.0035)
Observations = 2329	Parameters = 80	LL = -2992.0	AIC = 6144.1

Restricted Dynamic Estimates

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.3731 (0.3067, 0.4539)	36.58% (29.81%, 43.92%)	
2016	0.3795 (0.3025, 0.4761)	37.21% (29.20%, 45.99%)	
2017	0.4126 (0.3563, 0.4778)	40.27% (34.52%, 46.30%)	
2018	0.3836 (0.3271, 0.4497)	37.68% (31.79%, 43.97%)	
2019	0.3887 (0.3268, 0.4624)	38.14% (31.89%, 44.81%)	
Sigma			0.0020 (0.0013)
Observations = 2329 Parameters = 529 LL = -14131.0 AIC = 29319.9			

Restricted Matriculation-Dynamic Estimates

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.0071 (0.0026, 0.0194)	36.35% (29.47%, 43.84%)	
2016	0.0071 (0.0028, 0.0180)	38.25% (31.05%, 46.00%)	
2017	0.0105 (0.0041, 0.0267)	40.65% (35.15%, 46.39%)	
2018	0.0095 (0.0038, 0.0240)	36.59% (30.01%, 43.71%)	
2019	0.0095 (0.0035, 0.0255)	37.41% (31.04%, 44.26%)	
Sigma			0.0020** (0.0010)
Observations = 2329 Parameters = 568 LL = -14260.9 AIC = 29657.8			

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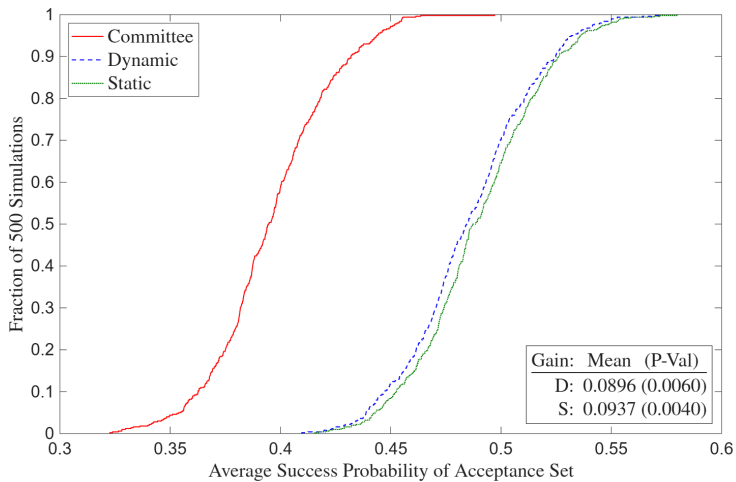
Deviation Gains Test

- Suppose committee adopts model's decision rule. Will average success probability of acceptance/matriculation set increase, and if so, by how much?
 - Solve model deterministically, compute average yearly success probability of model's acceptance/matriculation set, and compare to that of actual decision rule's set
 - $\hat{\theta}_S^R$ may be biased to rationalize suboptimal behavior, so plug in $\hat{\theta}_S^U$
- **Robustness:** Too few observations for holdout set $\Rightarrow$ implement randomization test
 - Compute average success probabilities with $B = 500$ perturbations $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$
 - P-value is fraction of perturbations with model's $\overline{P(\text{Success})} \leq$ committee's

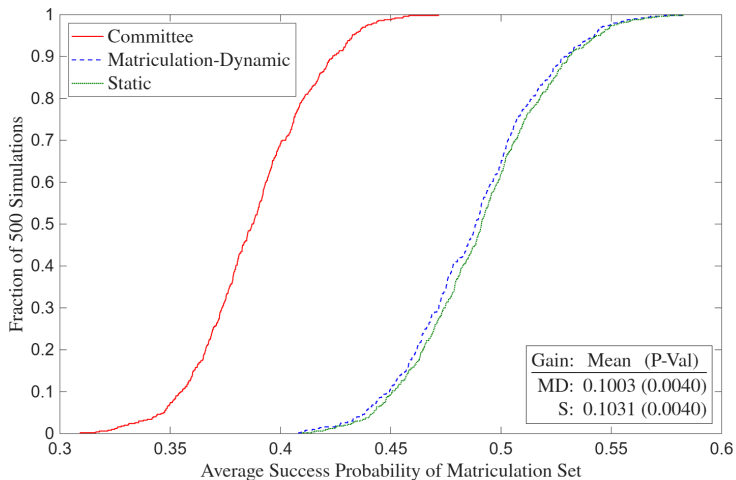
Theorem 3 (Proof)

For large B , the fraction of $\tilde{\theta}_{S,b}^U$ with gain $g(\tilde{\theta}_{S,b}^U) \leq 0$ is a p -value for $H_0: g_0 \leq 0$.

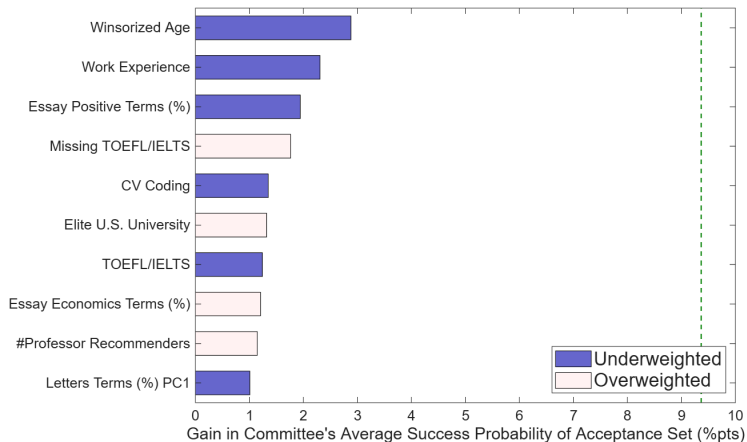
Deviation Gains



Deviation Gains (Matriculation)



Deviation Gains Attribution

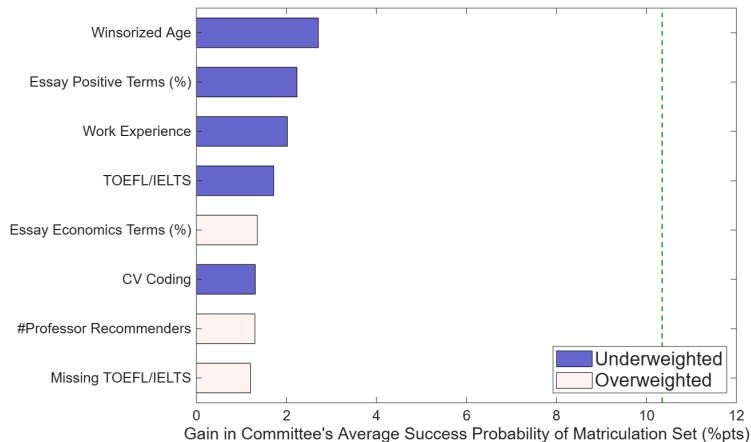


- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

» Full Deviation Gains Attribution (Matriculation)

» Deviation Gains Attribution Table (Matriculation)

Deviation Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

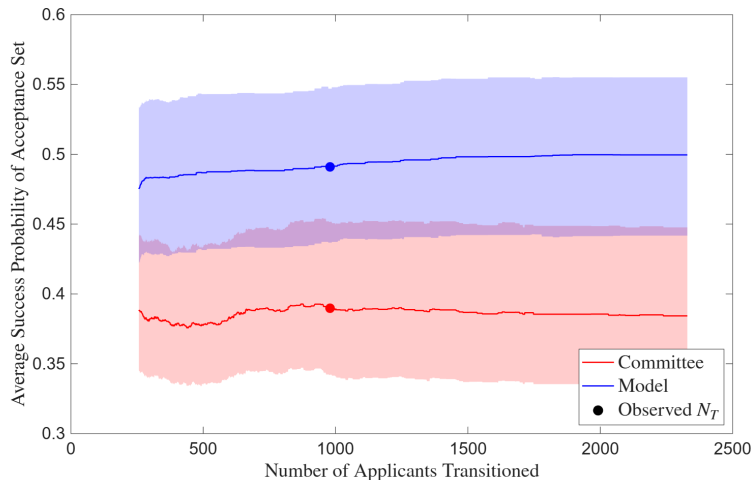
Regressions vs. Model

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
#Professor Recommenders	-0.0033	-0.0069
Missing TOEFL/IELTS	0.0070	-0.0008
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

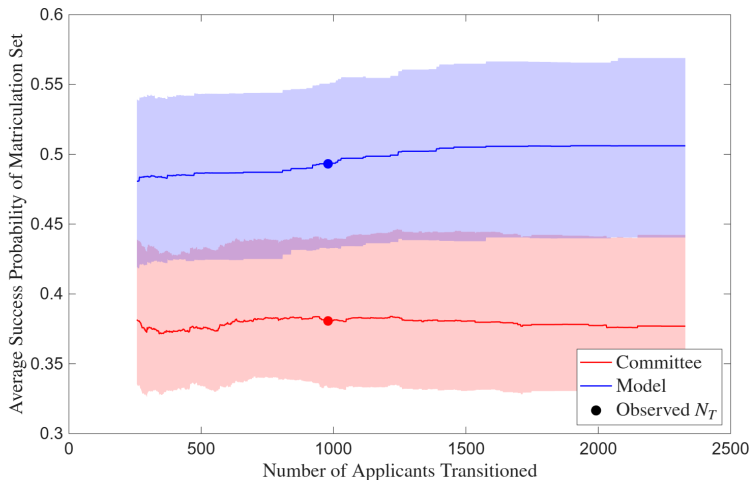
- Blue = underweighted, white = correctly weighted, red = overweighted
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

» Acceptance vs. Success AMEs

Benefit of Reading More Files



Benefit of Reading More Files (Matriculation)



Deviation Gains Interpretation

- Under meaningfully-sized deviation gains like those observed, committee may have incorrect beliefs about which types of applicants are most likely to succeed
- Randomization test is not a true out-of-sample test, and it cannot account for changes in the determinants of success over time, such as writing essays with AI
 - Attribution figures should be interpreted judiciously with that caveat in mind
- Committee may use success-predicting variables that I do not observe
 - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
 - Comprehensive data, especially committee score in z , protects against this limitation

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
 - ① For many applicants, estimated success probabilities align with those implied by committee's decisions. But committee misweights some applicant characteristics
 - ② Correctly weighting these characteristics would significantly increase admitted cohort's average success probability without reading more application files

» Simulated Results

» Waitlist Models

» Multiyear Models

Thank You!

A1: Synthetic Rows $\neq$ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	89	0	0	0	0	5	150
1	0	1	0	1	96	0	1	0	0	6	23
0	0	0	0	0	88	0	1	0	1	.	150
0	0	0	0	0	80	1	1	0	1	.	150
0	0	0	0	1	99	0	1	0	0	.	29
0	0	0	0	0	81	0	0	1	1	.	23
1	0	1	1	1	92	0	1	0	0	3	37
1	1	1	1	1	93	0	0	0	1	3	37
0	0	0	0	0	83	1	0	1	0	.	48
1	1	0	0	0	87	0	1	0	1	5	34



Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	77	0	0	0	1	.	30
0	0	0	0	1	89	0	0	0	0	.	25
0	0	0	0	0	85	1	0	0	0	.	150
0	0	0	0	0	76	0	1	0	1	.	51
1	1	0	0	0	88	0	1	0	0	8	150
1	0	1	0	0	88	0	0	0	1	5	150
1	1	0	0	0	84	0	0	1	0	2	20
0	0	0	0	0	82	0	1	0	0	.	35
0	0	0	0	0	70	1	0	0	0	.	150
1	1	0	0	1	93	0	1	0	0	5	57

A1: Dependent Variables

Table 1a: Dependent Variables

Admission	Success
1(Transition)	1(Complete Program)
<i>1(Accept)</i>	<i>1(Academic Placement)</i>
1(Waitlist)	1(Government Placement)
1(Open House)	1(Consulting Placement)
<i>1(Matriculate)</i>	<i>1(Tech Placement)</i>
1(Attend Program)	1(Other Placement)
Program Rank	1(No Placement)
	Placement Rank
	1(Top 100 Placement)

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
 - For non-tenure track, use weighted average of university's and maximum rankings
 - For non-academia, use average of ChatGPT, Claude, and Gemini's [rankings](#)

A1: Objective Variables

Table 1b: Objective Independent Variables

Demographic	Performance
Year	GRE Verbal Percentile
1 (<i>Covid Year</i>)	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
1 (Female)	Undergraduate GPA
1 (U.S.)	TOEFL/IELTS
1 (<i>U.S. African/Hispanic/Native</i>)	1 (Elite U.S. University)
1 (<i>U.S. Asian</i>)	1 (Elite U.S. Liberal Arts College)
1 (<i>U.S. Other</i>)	1 (Other U.S. University/College)
1 (International Asian)	1 (Elite Foreign University)
1 (International Other)	1 (Other Foreign University)
	1 (Graduate Degree)
	1 (Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	<i>Concentration Dummy Variables</i>

A1: Subjective Variables

Table 1c: Subjective Independent Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	#Math Courses
Letters Length	1(Advanced Math)
Positive Terms (%) (YS 2012)	CV Length
Negative Terms (%) (YS 2012)	CV Coding
Standout Terms (%) (BEMS 2022)	CV 1(Publication/Working Paper)
Ability Terms (%) (BEMS 2022)	Essay Length
Research Terms (%) (BEMS 2022)	Essay 1(Specific Professor)
Grindstone Terms (%) (BEMS 2022)	Essay Economics Terms (%)
Teaching Terms (%) (BEMS 2022)	Essay Research Terms (%)
Communal Terms (%) (BEMS 2022)	Essay Positive Terms (%)
Agentic Terms (%) (MHM 2009)	Essay Negative Terms (%)
	Field Dummy Variables

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Eonic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

A1: Dictionaries

● Recommendation Letters:

- **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop

● CVs and Essays:

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
 - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
 - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for “grindstone terms” (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
 - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
 - If GRE predicts success, committee should use it regardless of why

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
 - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
 - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
 - **Step 2:** Run OLS or heckprobit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
 - **Instruments:** Variables that predict matriculation but not the outcome

A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
 - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
 - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
 - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a “success” model
 - **Problem:** Not a test if admissions decisions are optimal
 - Need structural model of admissions to conclude about optimality

A1: Logistic Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$. Let $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$,

and $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}.$$

- $Y \in \{1(\text{Accept}), 1(\text{Attend Program}), 1(\text{Complete Program}), 1(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$. Then:
 $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

A1: Linear Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$, i.e. (Economics $\perp\!\!\!\perp$ Non-Economics) $|$ Predictors.

Let $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E)^2\right]$, and

$f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE})^2\right]$. Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2).$$

- $Y \in \{\mathbb{1}(\text{Standardized Committee Score}), \mathbb{1}(\text{Standardized Placement Rank})\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
Then $W = g(\hat{\beta}_1)'[\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$)

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
 - Need different skills at different stages in the program, such as grades vs. placement
 - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
 - **Admission:** May have different preparation and/or interests
 - **Completion:** Depending on country, may have fewer outside options
 - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
 - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
 - For economics, increased importance of elite predocs and Fed RAs

A2: Portfolio Choice to Cutoff Rule (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2.1: Portfolio Choice to Cutoff Rule with Shocks

Portfolio Choice Problem:

$$\begin{aligned} \max_{(d_1, \dots, d_N) \in \{A, R\}^N} & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + \epsilon_n(R) \mathbb{1}(d_n = R) \}, \\ \text{such that: } & \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \dots, d_N; \lambda) = \\ & \sum_{n=1}^N \{ [P(y_n = S | x_n, z_n) + \epsilon_n(A)] \mathbb{1}(d_n = A) + [\lambda + \epsilon_n(R)] \mathbb{1}(d_n = R) \} + \lambda(N_A - N) \end{aligned}$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \epsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S | x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 4 (Proof)

For large N , there is a cutoff rule solution with logistic CCPs.

► Model Framework

A2: Models of Admissions

- ① **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
 - ① **Static Model:** In Round 1, rank applicants based only on objective variables
 - ② **Myopic Model:** Like dynamic, but distribution of z not conditional on x
- ② **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)
- ③ **Waitlist-Dynamic/Hybrid Models:** In Round 1, rank applicants based on success probabilities. After Round 1, accept or waitlist applicants in Round 2. After acceptances matriculate or decline, accept or reject waitlisted applicants in Round 3. Favor likely matriculants in Round 2 to increase selectivity in Round 3

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \bar{k})$, where Success = $\mathbb{1}(\text{Complete Program})$ or $\mathbb{1}(\text{Good Placement})$. If reject applicant, get $P(y_2^* = S|t) = P(\text{Success})$ of year t 's marginal accepted applicant
- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P(\text{Success})$ of year t 's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
 - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ = expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
 - **Static:** If transition applicant, get $P(y = S|x, t, \bar{k})$, or $P(\text{Success})$ unconditional on z . It may differ from $P(y = S|x, z, t, \bar{k})$, though it still accounts for z 's effect on y
 - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$, replace $F(z|x, t)$ with $F(z|t)$

A2: Portfolio Choice to Cutoff Rule, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$

A2: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^* = S) \in [EV_{2,N_T+1}, EV_{2,N_T}]$

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)}$$

Success Probability:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P(y_2^* = S | t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

A2: Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t)\end{aligned}$$

- **Continuous:** $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x,t,j}}^2} (z_j - g(x, t)\beta_{F_{z|x,t,j}})^2 \right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}, \ell = 1, \dots, L_j - 1$

A2: Portfolio Choice to Cutoff Rule, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [P(y_n = S | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in [P(y_{N_T+1} = S | x_{N_T+1}), P(y_{N_T} = S | x_{N_T})]$

A2: Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F)}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_F)}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Success often unobservable for applicants from fewer than six years ago

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A)}_{\text{Conditional choice probabilities}} \\
 & \underbrace{(1 - P_{d_{2,n}}(\theta_A)) \mathbb{1}(d_{2,n}=R)}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^N \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{-c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{-c}|x, t} \\
 & \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{P_{y_n}(\theta_S) \mathbb{1}(y_n=S) (1 - P_{y_n}(\theta_S)) \mathbb{1}(y_n=F)}_{\text{Success probabilities}}
 \end{aligned}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

A2: Empirical Considerations

- **Selection Effects:** Track success for all applicants, including non-matriculants
 - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
 - In decision blocks containing success, fix *Program Rank* at this university's rank
 - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
 - **Example:** Recommendation letter has nuances that algorithms can't detect
 - Include subjective committee score in z as a proxy for this information

A2: Cutoff Rule with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^N P(m_n = M | w_n) \mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1, \dots, d_N; \lambda) = \sum_{n=1}^N [P(y_n = S | w_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] P(m_n = M | w_n) + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_n(w_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda^* \\ R & \text{if } P(y_n = S | w_n) < \lambda^* \end{cases}$$

A2: Cutoff Rule with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))]P(m = M|w)f(w)dw + \lambda(N_M - \overline{N}_M)$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

A2: Cutoff Rule with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | w_n) P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M | w_n) \mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} [P(y_n = S | w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M | w_n) + \lambda_2 (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | w_n) < \lambda_2^* \end{cases}$$

A2: Cutoff Rule with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n; \bar{\lambda}_2^*) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S | x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} P(m_n = M | x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n | x_n)$$

$$\mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) = \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

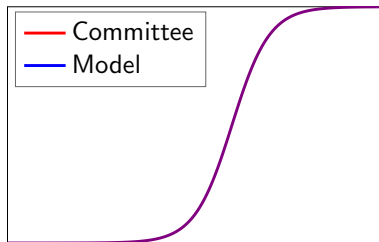
$$P_m = P(m = M | x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$$

$$\mathcal{L}_N^U(\theta) = \underbrace{\prod_{n=1}^{N_T} \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t}}_{\text{Density of } z|x, t} \underbrace{P_{d_{1,n}}(\theta_T) \left[P_{d_{2,n}}(\theta_A) P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)} \right]^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice and matriculation probabilities}}$$

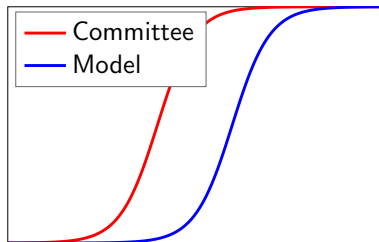
$$\underbrace{(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional Choice Probability}} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Outcome probabilities}} \prod_{n=N_T+1}^N \underbrace{f(z_n | x_n, t_n; \theta_F)}_{\text{Density of } z|x, t} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional Choice Probability}}$$

A2: How Deviation Gains Test Works

- 1 Write Bellman equations for committee's optimization problem
- 2 Estimate parameters. Rationality assumption distorts estimates
- 3 Re-estimate parameters without rationality, and use to solve model
- 4 Compare selections from model's decision rule to actual decision rule
- 5 To prevent overfitting, use perturbations of estimates for comparison



Optimal Committee



Suboptimal Committee

A2: Deviation Gains Algorithm 1

Dynamic/Static/Myopic:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \bar{k})$ for static. Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$, and accept top N_A survivors.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \bar{k})$ for static (best analogue). Transition top N_T , and reject the rest
- 2 Rank surviving applicants by $P(y = S|x, z, t, \bar{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until N_M matriculate.
Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of matriculation set, and compare to committee's set

A2: Deviation Gains Algorithm 2

Waitlist-Dynamic/Hybrid:

- 1 Rank each t 's applicants by $\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2)|x, t, \bar{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \bar{k})$ for hybrid. Transition top N_T , and reject the rest
- 2 Rank surviving applicants, with $\sigma = 0$ and $\beta = 1$, by $P(y = S|x, z, t, \bar{k}) - \beta \mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \bar{k}, \ddot{m}_2]$. Accept top N_{2A} , each matriculating with probability $P(m_2 = M|x, z, t)$, and waitlist the rest
- 3 Rank waitlisted applicants by $P(y = S|x, z, t, \bar{k})$ and accept them in descending order, each matriculating with probability $P(m_3 = M|x, z, t)$, until N_M matriculate. Compute $\overline{P(y = S|x, z, t, \bar{k})}$ of matriculation set, and compare to committee's set

A2: Randomization Test

Theorem 5 (Proof)

Let d^R be the model's decision rule, and d^U the actual decision rule. By optimality:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^R[P_{y_n}(\hat{\theta}_S^U)] = A\} \\ \geq \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^U[P_{y_n}(\hat{\theta}_A^U)] = A\} \end{aligned}$$

- But for perturbation $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$'s asymptotic distribution, can have:

$$\begin{aligned} \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^R[P_{y_n}(\hat{\theta}_S^U)] = A\} \\ < \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \tilde{\theta}_{S,b}^U) \mathbb{1}\{d_{1,n}[P_{y_n}(\hat{\theta}_U)] = T, d_{2,n}^U[P_{y_n}(\hat{\theta}_A^U)] = A\} \end{aligned}$$

- For matriculation and waitlist models, compare matriculants, not acceptances

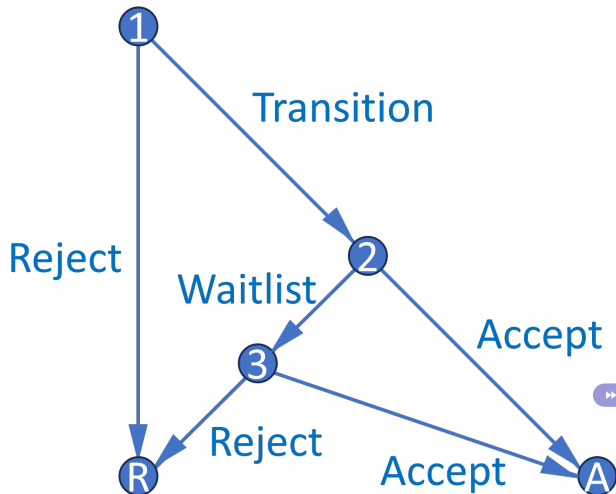
A2: Summary of Simulated Results

- **Simulated Results:** Estimation results are well behaved
 - **Dynamic/Static:** Deviation gains significantly positive for simulated dataset where committee misweights applicant characteristics, but not for optimal simulated dataset
 - **Matriculation-Dynamic/Static:** Same as above
 - **Waitlist-Dynamic/Hybrid:** Same as above, and matriculation estimates as expected
- Example of cutoff probability reversal if committee transitions few files in Round 1:
 - Three applicants: A, B, and C. Committee transitions two, accepts one
 - A is best and C is worst overall, but A is worst in objective variables
 - Committee rejects A in Round 1 and accepts B, who is worse than A
- By assuming h hours to read application file and w hourly wage, and calculating model's $P(\text{Success})$ induced by all possible N_T , can put dollar value on success
 - Alternatively, what if committee uses AI to read every applicant's file?

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
 - **Example:** Applicants with work experience more likely to succeed and matriculate
 - Can help committee find “people that we can afford” (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
 - Instead maximize combination of $P(\text{Success})$ and other objectives
 - **Example:** Committee may want diverse cohort (e.g. across fields):
 $P(\text{Success})^\alpha \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see Appendix 3)
 - Three-round models, and two-round models that repeat over infinite horizon
 - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

A3.1: Waitlist Decision Graph



- Three-round model features waitlist and accounts for applicants' decisions
- Dynamic mechanism incentivizes giving early acceptances to likely matriculants
- In practice, need several years of matriculation data for identification

» Conclusion

A3.1: Waitlist Model, Round 3

- Respecify Round 3 cutoff $P(y_3^* = S|t)$, where $t = \text{year}$, as $P(y_3^* = S|\overline{m}_{2A})$
 - $\overline{m}_{2A}$ = average early accepted applicant's matriculation probability
 - $P(y_3^* = S|\overline{m}_{2A})$ increases in $\overline{m}_{2A}$ because larger $\overline{m}_{2A}$ means fewer openings

$$v_3(x, z, t, \bar{k}, \overline{m}_{2A}, \epsilon_3) = \max_{d_3 \in \{A, R\}} u_3(x, z, t, \bar{k}, \overline{m}_{2A}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x, z, t, \bar{k}, \overline{m}_{2A}, d_3) = \underbrace{P(y = S|x, z, t, \bar{k})\mathbb{1}(d_3 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_3^* = S|\overline{m}_{2A})\mathbb{1}(d_3 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

A3.1: Waitlist Model, Round 2

- What information available at start of Round 2 determines $\overline{m_{2A}}$?
 - $\ddot{m}_2$ is defined for all transitioned applicants as $\overline{m_2}$ with m_2 excluded, where m_2 = applicant's own matriculation probability. Within a year, high m_2 implies low $\ddot{m}_2$
 - $\ddot{m}_2$ appears when applicant is waitlisted $\Rightarrow$ represents $\overline{m_2}$ of applicants committee *can* accept (vs. $\overline{m_{2A}}$, which represents $\overline{m_2}$ of applicants committee *does* accept)

$$v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2) = \max_{d_2 \in \{A, W\}} u_2(x, z, t, \bar{k}, \ddot{m}_2, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, \ddot{m}_2, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{\beta \mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2] \mathbb{1}(d_2 = W)}_{\text{If waitlist, get discounted } \mathbb{E}[\text{Social Surplus}]}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log \left(\sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \widetilde{\overline{m_{2A}}}, \tilde{d}_3) / \sigma} \right) dF_{\overline{m_{2A}} | \ddot{m}_2}(\widetilde{\overline{m_{2A}}} | \ddot{m}_2)$$

A3.1: Waitlist Dynamics

Theorem 6 (Proof)

Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.

- Compared to low m_2 applicant, high m_2 applicant has low $\ddot{m}_2$, and therefore low $\bar{F}_{\overline{m}_{2A}|\ddot{m}_2}$, $\mathbb{E}[P(\text{Cutoff})_3]$, and $\mathbb{E}[v_3]$. This increases relative payoff of accepting them
- Why not accept low m_2 applicant and fill spot with waitlisted high m_2 applicant?
 - ① m_2 declines between Rounds 2 and 3 and has farther to fall for high m_2 applicant
 - ② In general, matriculation probabilities can change unpredictably between rounds

★ **Upshot:** Accepting applicants who want to matriculate benefits the program

A3.1: Waitlist Model, Round 1

Round 1:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S | t) \mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2) | x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, W\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), \tilde{d}_2) / \sigma} \right) dF_{z|x, t}(\tilde{z} | x, t)$$

Conditional Choice Probabilities:

$$P(d_r | \text{states}) = \frac{\exp(u_r(\text{states}, d_r) / \sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r) / \sigma)}$$

A3.1: Waitlist Model, Round 1

Round 1 (Hybrid):

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{k}, d_1) = \underbrace{P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r)/\sigma)}$$

A3.1: Waitlist-Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, \ddot{m}_2; \theta_{A_2}) = \frac{\exp(f_{A_2}(x, z, \ddot{m}_2)\theta_{A_2})}{1 + \exp(f_{A_2}(x, z, \ddot{m}_2)\theta_{A_2})}$$

$$P_{d_3} = P(d_3 = A | x, z, \overline{m_{2A}}; \theta_{A_3}) = \frac{\exp(f_{A_3}(x, z, \overline{m_{2A}})\theta_{A_3})}{1 + \exp(f_{A_3}(x, z, \overline{m_{2A}})\theta_{A_3})}$$

Success/Matriculation Probabilities:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}, P_{m_2} = P(m_2 = M | x, z, t; \theta_{M_2}) = \frac{\exp(f_{M_2}(x, z, t)\theta_{M_2})}{1 + \exp(f_{M_2}(x, z, t)\theta_{M_2})}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P(y_3^* = S | \overline{m_{2A}}; \theta_{S_3^*}) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}$$

A3.1: Waitlist-Dynamic Parameterization 2

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), \tilde{d}_2)/\sigma} \right) dF_{z|x, t}(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_{J_z}} \left[\cdot \right] dF_{z|x, t, 1}(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_{J_z}, x, t) \cdots dF_{z|x, t, J_z}(\tilde{z}_{J_z}|x, t)\end{aligned}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \bar{k}, \ddot{m}_2] = \int_{\overline{m_{2A}}} \sigma \log \left(\sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \overline{m_{2A}}, \tilde{d}_3)/\sigma} \right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\overline{m_{2A}}|\ddot{m}_2)$$

- **Continuous:** $f_{z|x, t}(z_j|x, t; \beta_{F_{z|x, t, j}}, \sigma_{F_{z|x, t, j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x, t, j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x, t, j}}^2} (z_j - g(x, t)\beta_{F_{z|x, t, j}})^2 \right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x, t, j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x, t, j}})}{1 + \exp(g(x, t)\theta_{F_{z|x, t, j}})}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x, t, j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x, t, j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x, t, j}}^i)}, \ell = 1, \dots, L_j - 1$

A3.1: Waitlist-Dynamic Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z})}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{f(\overline{m_{2A,n}} | \ddot{m}_{2,n}; \theta_{F_m})}_{\text{Density of } \overline{m_{2A}} | \ddot{m}_2} \underbrace{P_{d_{1,n}}(\theta_T)}_{\text{Conditional choice probability}} \\
 & \underbrace{P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}}_{\text{Conditional choice probabilities}} \prod_{n=1}^{N_W} \underbrace{P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)} (1 - P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_{F_z})}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Estimate θ_{M_2} with 1st-stage likelihood to get $m_{2,n}$ = matriculation probability $P_{m_{2,n}}$:

$$\mathcal{L}_{N_T - N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)} (1 - P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$$

- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

A3.1: Waitlist Notes

- **Estimation:** Estimate $f_{\overline{m}_{2A}|\ddot{m}_2}$ on all transitioned applicants; slope parameter's expected sign is positive from variation in $\overline{m}_2$ across years (e.g. due to Covid)
 - **Assumption:** Effect of $\ddot{m}_2$ on $\overline{m}_{2A}$ is stationary across years, and therefore can be used to forecast changes in $\overline{m}_{2A}$ based on changes in $\ddot{m}_2$ from changes in d_2
- **Dynamics:** What if committee doesn't think dynamically?
 - If $\ddot{m}_2$ increasing increases $\overline{F}_{\overline{m}_{2A}|\ddot{m}_2}$, which increases $\mathbb{E}[P(\text{Cutoff})_3]$, then it should
- **Over/Undershooting:** Waitlist helps avoid both in reality, not just in expectation

A3.1: Symmetric Waitlist Model 1

Round 3:

$$v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3) = \max_{d_3 \in \{A, R\}} u_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, d_3) = P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_3 = A) + P(y_3^* = S | \overline{m_{2A}}, \overline{m_3}) \mathbb{1}(d_3 = R)$$

Round 2:

$$v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2) = \max_{d_2 \in \{A, W\}} u_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, d_2) =$$

$$P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A) + \beta \mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2, m_3] \mathbb{1}(d_2 = W), \text{ such that:}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \widetilde{\overline{m_3}}, \epsilon_3) | x, z, t, \bar{k}, \ddot{m}_2, m_3] =$$

$$\int_{\widetilde{\overline{m_{2A}}}} \int_{\widetilde{\overline{m_3}}} \sigma \log \left(\sum_{\widetilde{d_3} \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \widetilde{\overline{m_{2A}}}, \widetilde{\overline{m_3}}, \widetilde{d_3}) / \sigma} \right) dF_{\widetilde{\overline{m_3}} | m_3}(\widetilde{\overline{m_3}} | m_3) dF_{\overline{m_{2A}} | \ddot{m}_2}(\widetilde{\overline{m_{2A}}} | \ddot{m}_2)$$

A3.1: Symmetric Waitlist Model 2

Round 1:

$$v_1(x, t, \bar{k}) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

- **Hybrid:** $u_1(x, t, \bar{k}, d_1) = P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R)$

- **Dynamic:**

$$u_1(x, t, \bar{k}, d_1) = \mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2)|x, t, \bar{k}]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), m_3(x, \tilde{z}, t), \tilde{d}_2)/\sigma} \right) dF_{z|x, t}(\tilde{z}|x, t)$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r)/\sigma)}$$

A3.1: Symmetric Waitlist-Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, \ddot{m}_2, m_3; \theta_{A_2}) = \frac{\exp(f_{A_2}(x, z, \ddot{m}_2, m_3)\theta_{A_2})}{1 + \exp(f_{A_2}(x, z, \ddot{m}_2, m_3)\theta_{A_2})}$$

$$P_{d_3} = P(d_3 = A | x, z, \overline{m_{2A}}, \overline{m_3}; \theta_{A_3}) = \frac{\exp(f_{A_3}(x, z, \overline{m_{2A}}, \overline{m_3})\theta_{A_3})}{1 + \exp(f_{A_3}(x, z, \overline{m_{2A}}, \overline{m_3})\theta_{A_3})}$$

Success/Matriculation Probabilities:

$$P_y = P(y = S | x, z, t, k; \theta_S) = \frac{\exp(f_S(x, z, t, k)\theta_S)}{1 + \exp(f_S(x, z, t, k)\theta_S)}, P_{m_3} = P(m_3 = M | x, z, t; \theta_{M_3}) = \frac{\exp(f_{M_3}(x, z, t)\theta_{M_3})}{1 + \exp(f_{M_3}(x, z, t)\theta_{M_3})}$$

Cutoff Levels/Probabilities:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P(y_3^* = S | \overline{m_{2A}}, \overline{m_3}; \theta_{S_3^*}) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}}, \overline{m_3})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}}, \overline{m_3})\theta_{S_3^*})}$$

A3.1: Symmetric Waitlist-Dynamic Parameterization 2

★ Assume $(\overline{m_{2A}}, \overline{m_3})$ are conditionally independent:

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, m_3, \epsilon_2)|x, t, \bar{k}] &= \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{v_2(x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t), m_3(x, \tilde{z}, t), \tilde{d}_2)/\sigma} \right) dF_{z|x, t}(\tilde{z}|x, t) \\ &= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_{J_z}} [\cdot] dF_{z|x, t, 1}(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_{J_z}, x, t) \cdots dF_{z|x, t, J_z}(\tilde{z}_{J_z}|x, t)\end{aligned}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3)|x, z, t, \bar{k}, \ddot{m}_2, m_3] = \int_{\widetilde{\overline{m_{2A}}}} \int_{\widetilde{\overline{m_3}}} \sigma \log \left(\sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \widetilde{\overline{m_{2A}}}, \widetilde{\overline{m_3}}, \tilde{d}_3)/\sigma} \right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:** $f_{z|x, t}(z_j|x, t; \beta_{F_{z|x, t, j}}, \sigma_{F_{z|x, t, j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x, t, j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x, t, j}}^2} (z_j - g(x, t)\beta_{F_{z|x, t, j}})^2 \right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x, t, j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x, t, j}})}{1 + \exp(g(x, t)\theta_{F_{z|x, t, j}})}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x, t, j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x, t, j}}^\ell)}{1 + \sum_{i=1}^{L_j-1} \exp(g(x, t)\theta_{F_{z|x, t, j}}^i)}, \ell = 1, \dots, L_j - 1$

A3.1: Symmetric Waitlist-Dynamic Likelihood

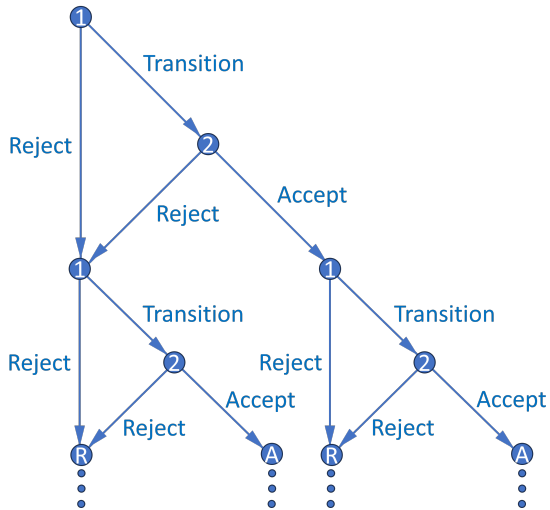
$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z})}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{f(\overline{m_{2A,n}} | \ddot{m}_{2,n}; \theta_{F_{m_2}})}_{\text{Density of } \overline{m_{2A}} | \ddot{m}_2} \underbrace{f(\overline{m_{3,n}} | m_{3,n}; \theta_{F_{m_3}})}_{\text{Density of } \overline{m_3} | m_3} \underbrace{P_{d_{1,n}}(\theta_T)}_{\text{Conditional choice probability}} \\
 & \underbrace{P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}}_{\text{Conditional choice probabilities}} \prod_{n=1}^{N_W} \underbrace{P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)} (1 - P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}}_{\text{Conditional choice probabilities}} \\
 & \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^N \underbrace{f(z_n^{\neg c} | x_n, t_n; \theta_{F_z})}_{\text{Density of } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Estimate θ_{M_3} with 1st-stage likelihood to get $m_{3,n}$ = matriculation probability $P_{m_{3,n}}$:

$$\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A, m_{3,n}=M)} (1 - P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A, m_{3,n}=D)}$$

- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

A3.2: Multiyear Decision Graph



A3.2: Multiyear Memory (Matriculation)

- Suppose committee maximizes discounted “lifetime” cohort success
- Respecify $P(y_2^* = S|t)$, where $t = \text{year}$, as $L(y_2^* = S|c_{-1})$
 - $c_{-1} = \text{previous year's cohort size}$. $L(\text{Cutoff}_2)$ increases in c_{-1} because committee must keep total number of students in program relatively constant across years
- What information available at start of Round 2 determines c ?
 - **If and only if $d_2 = A$:** $m = \text{applicant's matriculation probability}$
 - For simplicity, suppose matriculation probabilities are exogenous, and success probabilities depend only on applicant characteristics (i.e. not program rank)

A3.2: Multiyear Dynamics (Matriculation)

Theorem 7 (Proof)

Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.

- With memory, optimal dynamic decision may not equal optimal myopic one
 - High m means high $\bar{F}_{c|m}$, which means high $\mathbb{E}[L(\text{Cutoff}_2)']$ and future EV
 - Accepting likely matriculants above this year's margin means that next year in expectation, committee won't have to accept as many marginal applicants
- Harder to estimate since multiyear problem has no closed-form solution
 - Principal component analysis can help reduce dimensionality

A3.2: Multiyear Model (Matriculation), Round 2

$$v_2(x, z, t, m, c_{-1}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, m, c_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, m, c_{-1}, d_2) = \{P(y = S|x, z, t) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')|m]\} \mathbb{1}(d_2 = A) \\ + \{L(y_2^* = S|c_{-1}) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')]\} \mathbb{1}(d_2 = R), \text{ such that:}$$

$$\mathbb{E}[v_1(x', t', c, \epsilon_1')|(m)] = \int_c \int_{t'} \int_{x'} \sigma \log \left(\sum_{d_1' \in \{T, R\}} e^{u_1(x', t', c, d_1')/\sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c|(m))$$

Conditional Choice Probabilities:

$$P(d_2|x, z, t, m, c_{-1}) = \frac{\exp(u_2(x, z, t, m, c_{-1}, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(x, z, t, m, c_{-1}, \tilde{d}_2)/\sigma)}$$

A3.2: Multiyear Model (Matriculation), Round 1

$$v_1(x, t, c_{-1}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, c_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, c_{-1}, d_1) = \mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2) | x, t, c_{-1}] \mathbb{1}(d_1 = T) \\ + \{L(y_1^* = S | t) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')]\} \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2) | x, t, c_{-1}] = \\ \int_{\tilde{m}} \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \tilde{m}, c_{-1}, \tilde{d}_2) / \sigma} \right) dF_{z|x, t}(\tilde{z} | x, t) dF_{m|x, t}(\tilde{m} | x, t)$$

Conditional Choice Probabilities:

$$P(d_1 | x, t, c_{-1}) = \frac{\exp(u_1(x, t, c_{-1}, d_1) / \sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(x, t, c_{-1}, \tilde{d}_1) / \sigma)}$$

A3.2: Multiyear (Matriculation) Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t, c_{-1}; \theta_T) = \frac{\exp(f_T(x, t, c_{-1})\theta_T)}{1 + \exp(f_T(x, t, c_{-1})\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, m, c_{-1}; \theta_A) = \frac{\exp(f_A(x, z, m, c_{-1})\theta_A)}{1 + \exp(f_A(x, z, m, c_{-1})\theta_A)}$$

Success Probability:

$$P_y = P(y = S | x, z, t; \theta_S) = \frac{\exp(f_S(x, z, t)\theta_S)}{1 + \exp(f_S(x, z, t)\theta_S)}$$

Cutoff Levels:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = f_{S_1^*}(t)\theta_{S_1^*}$$
$$L_{y_2^*} = L(y_2^* = S | c_{-1}; \theta_{S_2^*}) = f_{S_2^*}(c_{-1})\theta_{S_2^*}$$

A3.2: Multiyear (Matriculation) Parameterization 2

- Assume (x, t) are independent, and next year is independent of current year:

$$\mathbb{E}[v_1(x', t', c, \epsilon_1') | (m)] = \int_c \int_{t'} \int_{x'} \sigma \log \left(\sum_{d_1' \in \{T, R\}} e^{u_1(x', t', c, d_1') / \sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c | (m)) =$$

$$\int_c \int_{t_1'} \cdots \int_{t_{J_t}'} \int_{x_1'} \cdots \int_{x_{J_x}'} \left[\cdot \right] dF_{x,1}(x_1' | x_2', \dots, x_{J_x}') \cdots dF_{x,J_x}(x_{J_x}') dF_{t,1}(t_1' | t_2', \dots, t_{J_t}') \cdots dF_{t,J_t}(t_{J_t}') dF_{c|(m)}(c | (m))$$

- Assume (z, m) are conditionally independent:

$$\mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2) | x, t, c_{-1}] = \int_{\tilde{m}} \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \tilde{m}, c_{-1}, \tilde{d}_2) / \sigma} \right) dF_{z|x,t}(\tilde{z} | x, t) dF_{m|x}(\tilde{m} | x, t) =$$

$$\int_{\tilde{m}} \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_{J_z}} \left[\cdot \right] dF_{z|x,t,1}(\tilde{z}_1 | \tilde{z}_2, \dots, \tilde{z}_{J_z}, x, t) \cdots dF_{z|x,t,J_z}(\tilde{z}_{J_z} | x, t) dF_{m|x,t}(\tilde{m} | x, t)$$

A3.2: Multiyear (Matriculation) Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \underbrace{\prod_{n=1}^{N_A} f(c_n | m_n; \theta_{F_c})}_{\text{Density of } c|m} \underbrace{\prod_{n=1}^{N_T} f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z})}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{f(m_n | x_n, t_n; \theta_{F_m})}_{\text{Density of } m|x, t} \\
 & \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A) (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\
 & \prod_{n=1}^N \underbrace{f(x_n; \theta_{F_x}) f(t_n; \theta_{F_t}) f(z_n^{\neg c} | x_n, t_n; \theta_{F_z})}_{\text{Densities of } x, t, \text{ and } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Estimate $f(c|m)$ on N_A accepted applicants only
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

A3.2: Multiyear Memory and Dynamics (Diversity)

- Let additional state a = any diversity variable affecting success
 - For example, $a = \mathbb{1}(\text{TheoryField})$; theory is less common than applied
 - Let $\bar{a}_{-1}$ = percentage of last year's acceptance set that's theory
 - If $|a - \bar{a}_{-1}|$ is small, $P(\text{Success})$ may decrease because future advisor may be too busy (effect may be unidentifiable if program never gets too imbalanced)

Theorem 8 (Proof)

Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z . Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p , the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.

- Optimal dynamic decision may not equal optimal myopic one. May accept diverse applicant with marginally lower $P(\text{Success})$ to increase $P(\text{Success})$ of next year's pool
 - Relative to $a = 0$, $a = 1 \Rightarrow$ higher $\bar{F}_{\bar{a}|a} \Rightarrow$ higher $\mathbb{E}[P(\text{Success})']$ and future EV

A3.2: Multiyear Model (Diversity), Round 2

$$v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, a, \bar{a}_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, a, \bar{a}_{-1}, d_2) = \{P(y = S|x, z, t, |a - \bar{a}_{-1}|) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a]\} \mathbb{1}(d_2 = A) \\ + \{L(y_2^* = S|t) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)]\} \mathbb{1}(d_2 = R), \text{ such that:}$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a] = \int_{\tilde{\bar{a}}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left(\sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{\bar{a}}, \tilde{t}', d'_1)/\sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\bar{a}|a}(\tilde{\bar{a}}|a)$$

Conditional Choice Probabilities:

$$P(d_2|x, z, t, a, \bar{a}_{-1}) = \frac{\exp(u_2(x, z, t, a, \bar{a}_{-1}, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(x, z, t, a, \bar{a}_{-1}, \tilde{d}_2)/\sigma)}$$

A3.2: Multiyear Model (Diversity), Round 1

$$v_1(x, t, \bar{a}_{-1}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{a}_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \bar{a}_{-1}, d_1) = \mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) | x, t, \bar{a}_{-1}] \mathbb{1}(d_1 = T) \\ + \{L(y_1^* = S | t) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)]\} \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) | x, t, \bar{a}_{-1}] = \int_{\tilde{a}} \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in A, R} e^{u_2(x, \tilde{z}, t, \tilde{a}, \bar{a}_{-1}, \tilde{d}_2) / \sigma} \right) dF_{z|x, t}(\tilde{z} | x, t) dF_a(\tilde{a})$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)] = \int_{\tilde{a}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left(\sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{t}', \tilde{a}, d'_1) / \sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\bar{a}}(\tilde{a})$$

Conditional Choice Probabilities:

$$P(d_1 | x, t, \bar{a}_{-1}) = \frac{\exp(u_1(x, t, \bar{a}_{-1}, d_1) / \sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(x, t, \bar{a}_{-1}, \tilde{d}_1) / \sigma)}$$

A3.2: Multiyear (Diversity) Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P(d_1 = T | x, t, \bar{a}_{-1}; \theta_T) = \frac{\exp(f_T(x, t, \bar{a}_{-1})\theta_T)}{1 + \exp(f_T(x, t, \bar{a}_{-1})\theta_T)}$$

$$P_{d_2} = P(d_2 = A | x, z, t, a, \bar{a}_{-1}; \theta_A) = \frac{\exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}{1 + \exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}$$

Success Probability:

$$P_y = P(y = S | x, z, t, |a - \bar{a}_{-1}|; \theta_S) = \frac{\exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}{1 + \exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}$$

Cutoff Levels:

$$L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*}) = f_{S_1^*}(t)\theta_{S_1^*}$$

$$L_{y_2^*} = L(y_2^* = S | t; \theta_{S_2^*}) = f_{S_2^*}(t)\theta_{S_2^*}$$

A3.2: Multiyear (Diversity) Parameterization 2

- Assume $(x, t, \bar{a})$ mutually independent, such that $\bar{a}|a$:

$$\begin{aligned} \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a] &= \int_{\tilde{\bar{a}}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left(\sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{t}', \tilde{\bar{a}}, d'_1)/\sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\bar{a}|a}(\tilde{\bar{a}}|a) = \\ &\int_{\tilde{\bar{a}}} \int_{\tilde{t}'_1} \cdots \int_{\tilde{t}'_{J_t}} \int_{\tilde{x}'_1} \cdots \int_{\tilde{x}'_{J_x}} \left[\cdot \right] F_{x,1}(x'_1|x'_2, \dots, x'_{J_x}) \cdots dF_{x,J_x}(x'_{J_x}) dF_{t,1}(t'_1|t'_2, \dots, t'_{J_t}) \cdots dF_{t,J_t}(t'_{J_t}) dF_{\bar{a}|a}(\tilde{\bar{a}}|a) \end{aligned}$$

- Assume (z, a) independent, such that $z|x$:

$$\begin{aligned} \mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2)|x, t, \bar{a}_{-1}] &= \int_{\tilde{\bar{a}}} \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in A, R} e^{u_2(x, \tilde{z}, t, \tilde{\bar{a}}, \bar{a}_{-1}, \tilde{d}_2)/\sigma} \right) dF_{z|x,t}(\tilde{z}|x, t) dF_a(\tilde{\bar{a}}) \\ &= \int_{\tilde{\bar{a}}} \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t) dF_a(\tilde{\bar{a}}) \end{aligned}$$

A3.2: Multiyear (Diversity) Likelihood

$$\begin{aligned}
 \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_A} \underbrace{f(\bar{a}_n | a_n; \theta_{F_{\bar{a}}})}_{\text{Density of } \bar{a}|a} \prod_{n=1}^{N_T} \underbrace{f(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z})}_{\text{Density of } z^c | z^{\neg c}, x, t} \underbrace{f(a_n; \theta_{F_a})}_{\text{Density of } a} \\
 & \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^N \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \\
 & \prod_{n=1}^N \underbrace{f(x_n; \theta_{F_x}) f(t_n; \theta_{F_t}) f(z_n^{\neg c} | x_n, t_n; \theta_{F_z})}_{\text{Densities of } x, t, \text{ and } z^{\neg c} | x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}
 \end{aligned}$$

- Estimate $f(\bar{a}|a)$ on N_A accepted applicants only
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

A4: Economics Department Summary Statistics

Table 2: Economics Department Summary Statistics

(a) Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1820	0.3973	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6387	0.3026	3.6645	0.3023	3.7227	0.2610	3.7068	0.2880
TOEFL iELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Prolific Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL iELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

(b) Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2031
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

Table 3: Economics Department Correlation Matrix

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A4: Economics Department Rec Letters Principal Components

Table 4: Economics Department Rec Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

A4: Economics Department Logistic Regression Results

Table 5a: Economics Department Logistic Regression Results

[illegible]

- **Accept** ($p < .1$): 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success** ($p < .1$): Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A4: Economics Department Logistic Regression AMEs

Table 5b: Economics Department Logistic Regression AMEs

Dependent Variable	Intercept	Age	Age ²	Age ³	Female	International Asian	GRE Quantitative Percentile	GRE Analytical Percentile	TOEFL/IELTS	Work Experience	Committee Score	CV Coding	CV Publishing/Working Paper	Essay Rank	Program Rank	Success Rank
2017	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2018	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2019	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Female	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
U.S.	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
International Asian	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
GRE Quantitative Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
GRE Analytical Percentile	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
TOEFL/IELTS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Work Experience	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Committee Score	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CV Coding	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CV Publishing/Working Paper	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Essay Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Success Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

- **Accept** ($p < .1$): 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success** ($p < .1$): Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

A4: Economics Department Logistic Regression Results (Synthetic)

Table 6a: Economics Department Logistic Regression Results (Synthetic)

Explanatory Variable	Intercept	Age	Age ²	Female	Elite U.S. University	Elite Foreign University	Missing Undergraduate Institution	Graduate Degree	High Experiences	#Public Recommendation	Missing GRE	Missing TOEFL/IELTS	Committee Score	#Math Courses	Admitted Status	Letter Length/100	CV Length	CV Coding	CV Publications/Working Paper	Research Paper	Research Paper Rank	Program Rank	Missing Program Rank
Intercept	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Age ²	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Female	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Elite U.S. University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Elite Foreign University	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Missing Undergraduate Institution	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Graduate Degree	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
High Experiences	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
#Public Recommendation	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Missing GRE	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Missing TOEFL/IELTS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Committee Score	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
#Math Courses	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Admitted Status	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Letter Length/100	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CV Length	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CV Coding	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CV Publications/Working Paper	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Research Paper	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Research Paper Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Missing Program Rank	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

- **Accept** ($p < .1$): 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Program Committee Score (-)
- **Success** ($p < .1$): Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A4: Economics Department Logistic Regression AMEs (Synthetic)

Table 6b: Economics Department Logistic Regression AMEs (Synthetic)

Variable	Mean	SD	Min	Max	Skewness	Kurtosis	Shapiro-Wilk	Normality
2015	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2017	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2018	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2019	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2020	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2021	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2022	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2023	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2024	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2025	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2026	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2027	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2028	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2029	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2030	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2031	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2032	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2033	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2034	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2035	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2036	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2037	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2038	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2039	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2040	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2041	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2042	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2043	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2044	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2045	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2046	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2047	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2048	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2049	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2050	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2051	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2052	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2053	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2054	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2055	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2056	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2057	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2058	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2059	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2060	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2061	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2062	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2063	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2064	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2065	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2066	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2067	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2068	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2069	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2070	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2071	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2072	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2073	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2074	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2075	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2076	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2077	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2078	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2079	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2080	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2081	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2082	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2083	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2084	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2085	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2086	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2087	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2088	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2089	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2090	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2091	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2092	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2093	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2094	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2095	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2096	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2097	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2098	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2099	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2100	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2101	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2102	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2103	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2104	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2105	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2106	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2107	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2108	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2109	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2110	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2111	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2112	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2113	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2114	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2115	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2116	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2117	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2118	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2119	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2120	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2121	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2122	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2123	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2124	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2125	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2126	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2127	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2128	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2129	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2130	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2131	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2132	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2133	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2134	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2135	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2136	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2137	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2138	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2139	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2140	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2141	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2142	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2143	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2144	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2145	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2146	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2147	-0.000	0.000	-0.000	0.000	0.000	0.000	0.999	0.000
2148	-0.000	0.000	-0.000					

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

A4: Economics Department Static Estimates

Table 7: Economics Department Static Estimates

(a) Unrestricted

(b) Restricted

Category	Transition		Asympt		Summe 1		Summe 2		Summe 3	Summe 4	Summe 5	Summe 6	Summe 7	Summe 8	Summe 9	Summe 10	Summe 11	Summe 12	Summe 13	Summe 14	Summe 15	Summe 16	Summe 17	Summe 18	Summe 19	Summe 20	Summe 21	Summe 22	Summe 23	Summe 24	Summe 25	Summe 26	Summe 27	Summe 28	Summe 29	Summe 30	Summe 31	Summe 32	Summe 33	Summe 34	Summe 35	Summe 36	Summe 37	Summe 38	Summe 39	Summe 40	Summe 41	Summe 42	Summe 43	Summe 44	Summe 45	Summe 46	Summe 47	Summe 48	Summe 49	Summe 50	Summe 51	Summe 52	Summe 53	Summe 54	Summe 55	Summe 56	Summe 57	Summe 58	Summe 59	Summe 60	Summe 61	Summe 62	Summe 63	Summe 64	Summe 65	Summe 66	Summe 67	Summe 68	Summe 69	Summe 70	Summe 71	Summe 72	Summe 73	Summe 74	Summe 75	Summe 76	Summe 77	Summe 78	Summe 79	Summe 80	Summe 81	Summe 82	Summe 83	Summe 84	Summe 85	Summe 86	Summe 87	Summe 88	Summe 89	Summe 90	Summe 91	Summe 92	Summe 93	Summe 94	Summe 95	Summe 96	Summe 97	Summe 98	Summe 99	Summe 100	Summe 101	Summe 102	Summe 103	Summe 104	Summe 105	Summe 106	Summe 107	Summe 108	Summe 109	Summe 110	Summe 111	Summe 112	Summe 113	Summe 114	Summe 115	Summe 116	Summe 117	Summe 118	Summe 119	Summe 120	Summe 121	Summe 122	Summe 123	Summe 124	Summe 125	Summe 126	Summe 127	Summe 128	Summe 129	Summe 130	Summe 131	Summe 132	Summe 133	Summe 134	Summe 135	Summe 136	Summe 137	Summe 138	Summe 139	Summe 140	Summe 141	Summe 142	Summe 143	Summe 144	Summe 145	Summe 146	Summe 147	Summe 148	Summe 149	Summe 150	Summe 151	Summe 152	Summe 153	Summe 154	Summe 155	Summe 156	Summe 157	Summe 158	Summe 159	Summe 160	Summe 161	Summe 162	Summe 163	Summe 164	Summe 165	Summe 166	Summe 167	Summe 168	Summe 169	Summe 170	Summe 171	Summe 172	Summe 173	Summe 174	Summe 175	Summe 176	Summe 177	Summe 178	Summe 179	Summe 180	Summe 181	Summe 182	Summe 183	Summe 184	Summe 185	Summe 186	Summe 187	Summe 188	Summe 189	Summe 190	Summe 191	Summe 192	Summe 193	Summe 194	Summe 195	Summe 196	Summe 197	Summe 198	Summe 199	Summe 200	Summe 201	Summe 202	Summe 203	Summe 204	Summe 205	Summe 206	Summe 207	Summe 208	Summe 209	Summe 210	Summe 211	Summe 212	Summe 213	Summe 214	Summe 215	Summe 216	Summe 217	Summe 218	Summe 219	Summe 220	Summe 221	Summe 222	Summe 223	Summe 224	Summe 225	Summe 226	Summe 227	Summe 228	Summe 229	Summe 230	Summe 231	Summe 232	Summe 233	Summe 234	Summe 235	Summe 236	Summe 237	Summe 238	Summe 239	Summe 240	Summe 241	Summe 242	Summe 243	Summe 244	Summe 245	Summe 246	Summe 247	Summe 248	Summe 249	Summe 250	Summe 251	Summe 252	Summe 253	Summe 254	Summe 255	Summe 256	Summe 257	Summe 258	Summe 259	Summe 260	Summe 261	Summe 262	Summe 263	Summe 264	Summe 265	Summe 266	Summe 267	Summe 268	Summe 269	Summe 270	Summe 271	Summe 272	Summe 273	Summe 274	Summe 275	Summe 276	Summe 277	Summe 278	Summe 279	Summe 280	Summe 281	Summe 282	Summe 283	Summe 284	Summe 285	Summe 286	Summe 287	Summe 288	Summe 289	Summe 290	Summe 291	Summe 292	Summe 293	Summe 294	Summe 295	Summe 296	Summe 297	Summe 298	Summe 299	Summe 300	Summe 301	Summe 302	Summe 303	Summe 304	Summe 305	Summe 306	Summe 307	Summe 308	Summe 309	Summe 310	Summe 311	Summe 312	Summe 313	Summe 314	Summe 315	Summe 316	Summe 317	Summe 318	Summe 319	Summe 320	Summe 321	Summe 322	Summe 323	Summe 324	Summe 325	Summe 326	Summe 327	Summe 328	Summe 329	Summe 330	Summe 331	Summe 332	Summe 333	Summe 334	Summe 335	Summe 336	Summe 337	Summe 338	Summe 339	Summe 340	Summe 341	Summe 342	Summe 343	Summe 344	Summe 345	Summe 346	Summe 347	Summe 348	Summe 349	Summe 350	Summe 351	Summe 352	Summe 353	Summe 354	Summe 355	Summe 356	Summe 357	Summe 358	Summe 359	Summe 360	Summe 361	Summe 362	Summe 363	Summe 364	Summe 365	Summe 366	Summe 367	Summe 368	Summe 369	Summe 370	Summe 371	Summe 372	Summe 373	Summe 374	Summe 375	Summe 376	Summe 377	Summe 378	Summe 379	Summe 380	Summe 381	Summe 382	Summe 383	Summe 384	Summe 385	Summe 386	Summe 387	Summe 388	Summe 389	Summe 390	Summe 391	Summe 392	Summe 393	Summe 394	Summe 395	Summe 396	Summe 397	Summe 398	Summe 399	Summe 400	Summe 401	Summe 402	Summe 403	Summe 404	Summe 405	Summe 406	Summe 407	Summe 408	Summe 409	Summe 410	Summe 411	Summe 412	Summe 413	Summe 414	Summe 415	Summe 416	Summe 417	Summe 418	Summe 419	Summe 420	Summe 421	Summe 422	Summe 423	Summe 424	Summe 425	Summe 426	Summe 427	Summe 428	Summe 429	Summe 430	Summe 431	Summe 432	Summe 433	Summe 434	Summe 435	Summe 436	Summe 437	Summe 438	Summe 439	Summe 440	Summe 441	Summe 442	Summe 443	Summe 444	Summe 445	Summe 446	Summe 447	Summe 448	Summe 449	Summe 450	Summe 451	Summe 452	Summe 453	Summe 454	Summe 455	Summe 456	Summe 457	Summe 458	Summe 459	Summe 460	Summe 461	Summe 462	Summe 463	Summe 464	Summe 465	Summe 466	Summe 467	Summe 468	Summe 469	Summe 470	Summe 471	Summe 472	Summe 473	Summe 474	Summe 475	Summe 476	Summe 477	Summe 478	Summe 479	Summe 480	Summe 481	Summe 482	Summe 483	Summe 484	Summe 485	Summe 486	Summe 487	Summe 488	Summe 489	Summe 490	Summe 491	Summe 492	Summe 493	Summe 494	Summe 495	Summe 496	Summe 497	Summe 498	Summe 499	Summe 500	Summe 501	Summe 502	Summe 503	Summe 504	Summe 505	Summe 506	Summe 507	Summe 508	Summe 509	Summe 510	Summe 511	Summe 512	Summe 513	Summe 514	Summe 515	Summe 516	Summe 517	Summe 518	Summe 519	Summe 520	Summe 521	Summe 522	Summe 523	Summe 524	Summe 525	Summe 526	Summe 527	Summe 528	Summe 529	Summe 530	Summe 531	Summe 532	Summe 533	Summe 534	Summe 535	Summe 536	Summe 537	Summe 538	Summe 539	Summe 540	Summe 541	Summe 542	Summe 543	Summe 544	Summe 545	Summe 546	Summe 547	Summe 548	Summe 549	Summe 550	Summe 551	Summe 552	Summe 553	Summe 554	Summe 555	Summe 556	Summe 557	Summe 558	Summe 559	Summe 560	Summe 561	Summe 562	Summe 563	Summe 564	Summe 565	Summe 566	Summe 567	Summe 568	Summe 569	Summe 570	Summe 571	Summe 572	Summe 573	Summe 574	Summe 575	Summe 576	Summe 577	Summe 578	Summe 579	Summe 580	Summe 581	Summe 582	Summe 583	Summe 584	Summe 585	Summe 586	Summe 587	Summe 588	Summe 589	Summe 590	Summe 591	Summe 592	Summe 593	Summe 594	Summe 595	Summe 596	Summe 597	Summe 598	Summe 599	Summe 600	Summe 601	Summe 602	Summe 603	Summe 604	Summe 605	Summe 606	Summe 607	Summe 608	Summe 609	Summe 610	Summe 611	Summe 612	Summe 613	Summe 614	Summe 615	Summe 616	Summe 617	Summe 618	Summe 619	Summe 620	Summe 621	Summe 622	Summe 623	Summe 624	Summe 625	Summe 626	Summe 627	Summe 628	Summe 629	Summe 630	Summe 631	Summe 632	Summe 633	Summe 634	Summe 635	Summe 636	Summe 637	Summe 638	Summe 639	Summe 640	Summe 641	Summe 642	Summe 643	Summe 644	Summe 645	Summe 646	Summe 647	Summe 648	Summe 649	Summe 650	Summe 651	Summe 652	Summe 653	Summe 654	Summe 655	Summe 656	Summe 657	Summe 658	Summe 659	Summe 660	Summe 661	Summe 662	Summe 663	Summe 664	Summe 665	Summe 666	Summe 667	Summe 668	Summe 669	Summe 670	Summe 671	Summe 672	Summe 673	Summe 674	Summe 675	Summe 676	Summe 677	Summe 678	Summe 679	Summe 680	Summe 681	Summe 682	Summe 683	Summe 684	Summe 685	Summe 686	Summe 687	Summe 688	Summe 689	Summe 690	Summe 691	Summe 692	Summe 693	Summe 694	Summe 695	Summe 696	Summe 697	Summe 698	Summe 699	Summe 700	Summe 701	Summe 702	Summe 703	Summe 704	Summe 705	Summe 706	Summe 707	Summe 708	Summe 709	Summe 710	Summe 711	Summe 712	Summe 713	Summe 714	Summe 715	Summe 716	Summe 717	Summe 718	Summe 719	Summe 720	Summe 721	Summe 722	Summe 723	Summe 724	Summe 725	Summe 726	Summe 727	Summe 728	Summe 729	Summe 730	Summe 731	Summe 732	Summe 733	Summe 734	Summe 735	Summe 736	Summe 737	Summe 738	Summe 739	Summe 740	Summe 741	Summe 742	Summe 743	Summe 744	Summe 745	Summe 746	Summe 747	Summe 748	Summe 749	Summe 750	Summe 751	Summe 752	Summe 753	Summe 754	Summe 755	Summe 756	Summe 757	Summe 758	Summe 759	Summe 760	Summe 761	Summe 762	Summe 763	Summe 764	Summe 765	Summe 766	Summe 767	Summe 768	Summe 769	Summe 770	Summe 771	Summe 772	Summe 773	Summe 774	Summe 775	Summe 776	Summe 777	Summe 778	Summe 779	Summe 780	Summe 781	Summe 782	Summe 783	Summe 784	Summe 785	Summe 786	Summe 787	Summe 788	Summe 789	Summe 790	Summe 791	Summe 792	Summe 793	Summe 794	Summe 795	Summe 796	Summe 797	Summe 798	Summe 799	Summe 800	Summe 801	Summe 802	Summe 803	Summe 804	Summe 805	Summe 806	Summe 807	Summe 808	Summe 809	Summe 810	Summe 811	Summe 812	Summe 813	Summe 814	Summe 815	Summe 816	Summe 817	Summe 818	Summe 819	Summe 820	Summe 821	Summe 822	Summe 823	Summe 824	Summe 825	Summe 826	Summe 827	Summe 828	Summe 829	Summe 830	Summe 831	Summe 832	Summe 833	Summe 834	Summe 835	Summe 836	Summe 837	Summe 838	Summe 839	Summe 840	Summe 841	Summe 842	Summe 843	Summe 844	Summe 845	Summe 846	Summe 847	Summe 848	Summe 849	Summe 850	Summe 851	Summe 852	Summe 853	Summe 854	Summe 855	Summe 856	Summe 857	Summe 858	Summe 859	Summe 860	Summe 861	Summe 862	Summe 863	Summe 864	Summe 865	Summe 866	Summe 867	Summe 868	Summe 869	Summe 870	Summe 871	Summe 872	Summe 873	Summe 874	Summe 875	Summe 876	Summe 877	Summe 878	Summe 879	Summe 880	Summe 881	Summe 882	Summe 883	Summe 884	Summe 885	Summe 886	Summe 887	Summe 888	Summe 889	Summe 890
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►► Restricted Static Estimates

A4: Economics Department Dynamic Estimates

Table 8: Economics Department Dynamic Estimates

[illegible][illegible]

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[illegible]

A4: Economics Department Myopic Estimates

Table 9: Economics Department Myopic Estimates

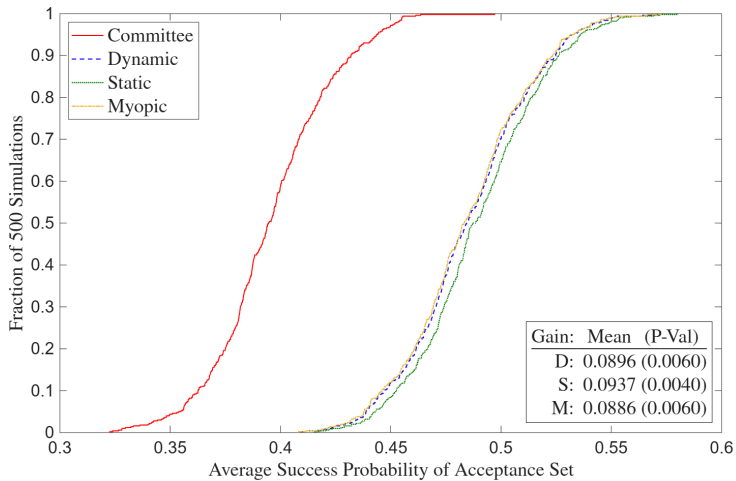
	Confidence		Dissidence		Anger		Sadness	
	Confidence	Confidence	Confidence	Confidence	Confidence	Confidence	Confidence	
China	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
2016	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
2017	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
2018	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
2019	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
United States	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Financial	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Worldwide	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
U.S.	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
International	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF Quantitative Prompts	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF Qualitative Prompts	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
TSPS17S17S17	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
U.S. State of Affairs	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Elie Wiesel Nobel Lecture	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Other Foreign Language	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Other Translations	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Other Foreign-Language Institutions	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Geological Events	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
World Empires	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
(F)Public Recommendations	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
(F)Public Recommendations	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Using CRF	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Using TSPS17S17S17	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Common Scores	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
(F)Public Comments	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Advanced Math	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Length (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Length (Length) (FC)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
Length (Length) (FC)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
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CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0.013721	0.013721	
CRF (Length)	0.986279	0.986279	0.013721	0.013721	0.013721	0		

[illegible]

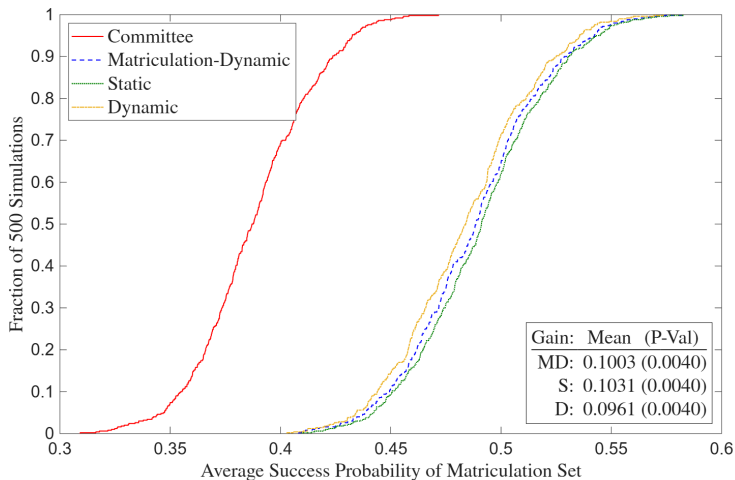
Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%

Obs Forum LL AIC	3326	267	-36678.7	268.8
Robust standard errors in parentheses. Two-tail tests: *0.10, **0.05, ***0.01				

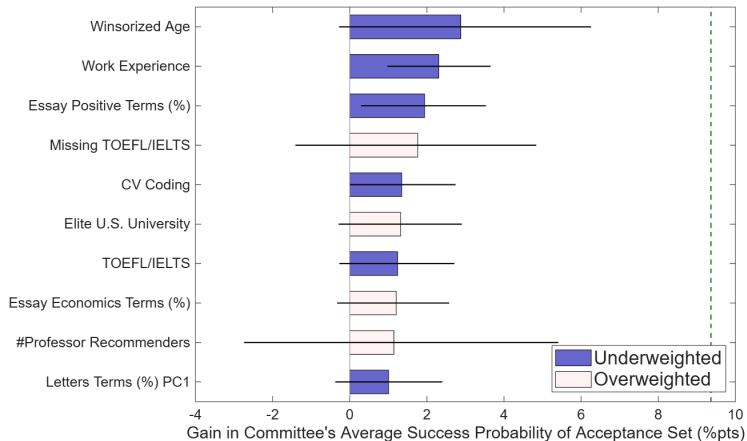
A4: Economics Department Deviation Gains



A4: Economics Department Deviation Gains (Matriculation)

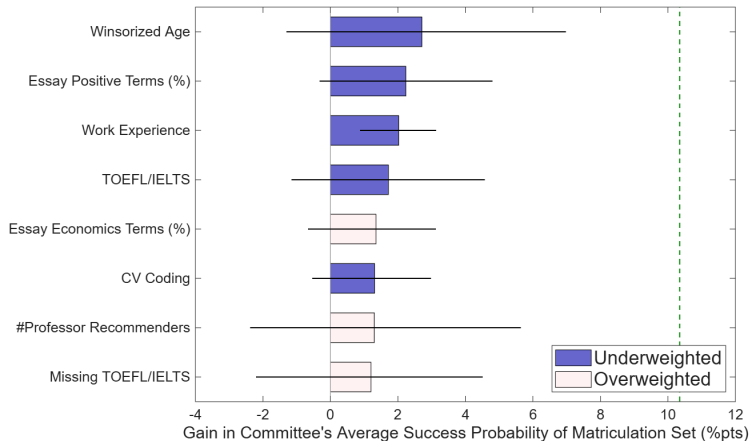


A4: Economics Department Gains Attribution



- 1 Experience
- 2 English Proficiency
- 3 Essay/Letters Subtext
- 4 Industry Background
- 5 Undergraduate

A4: Economics Department Gains Attribution (Matriculation)



- 1 Experience
- 2 English Proficiency
- 3 Essay Subtext
- 4 Industry Background

A4: Economics Department Static Attribution

Table 11: Economics Department Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.8788	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.3065	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.9410	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.7635	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.3478	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Elite U.S. University	1.3179	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
TOEFL/IELTS	1.2400	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294
Essay Economics Terms (%)	1.2043	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
#Professor Recommenders	1.1455	2.6230	2.6148	1.3852	2.1362	-1.6680	-0.6493
Letters Terms (%) PC1	1.0074	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018

►► Deviation Gains Attribution

A4: Economics Department Matriculation-Static Attribution

Table 12: Economics Department Matriculation-Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.7099	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	2.2339	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	2.0220	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.7179	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.3503	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.3100	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374
#Professor Recommenders	1.3005	2.6230	2.6171	1.4173	2.2255	-1.6321	-0.5312
Missing TOEFL/IELTS	1.1999	0.6076	0.6858	0.0442	0.4368	-1.3075	-0.5099

►► Deviation Gains Attribution (Matriculation)

A4: Graduate School Means by Program

Table 13a: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Means by Program, Accept = 1

Table 13b: Graduate School Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

A4: Graduate School Correlation Matrix

Table 14: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREa	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

A4: Graduate School Logistic Regression Results

Table 15a: Graduate School Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966** (0.2010)	-0.4332* (0.2612)	-0.5792** (0.3470)	-0.5501 (0.4120)	-0.1766 (0.5545)
Winsorized Age	-0.0995*** (0.0358)	0.0454* (0.0271)	0.0380 (0.0327)	0.0222 (0.0532)	-0.0520 (0.0890)
Female	0.4079** (0.1681)	0.6721*** (0.2299)	0.6539** (0.2971)	0.1311 (0.3452)	-0.5660 (0.5606)
U.S. African/Hispanic/Native	0.4221 (0.5258)	1.0081*** (0.3545)	1.1601*** (0.4426)	0.0120 (0.8411)	-0.8263 (1.1511)
U.S. Asian	-0.5394 (0.4190)	-0.5660 (0.6504)	-0.3260 (1.2512)	0.8638 (0.7201)	1.1711 (0.7276)
International Asian	-0.8445*** (0.2945)	-0.1853 (0.4146)	0.9405 (0.7233)	0.4890 (0.6720)	-0.7827 (0.8323)
International Other	0.5034 (0.3172)	0.3781 (0.4131)	1.3529*** (0.5086)	-0.3061 (0.7483)	0.6239 (1.1070)
GRE Verbal Percentile	0.0155*** (0.0050)	0.0281* (0.0160)	0.0753** (0.0344)	-0.0092 (0.0154)	0.9607*** (0.0218)
GRE Quantitative Percentile	0.1547*** (0.0186)	0.0161* (0.0086)	0.0219* (0.0116)	0.0211 (0.0150)	-0.0138 (0.0159)
GRE Analytic Percentile	0.0179*** (0.0042)	0.0199** (0.0094)	0.0030 (0.0134)	0.0175 (0.0127)	-0.0202 (0.0218)
Undergraduate GPA	2.2212*** (0.7219)	0.6969 (0.5476)	1.3849** (0.5513)	1.2646 (0.9703)	1.7722* (0.9487)
TOEFL/IELTS	0.3878 (0.2883)	0.6456 (0.4708)	1.0963** (0.5526)	0.9655* (0.5496)	-0.4656 (0.8112)
Elite U.S. University	0.8850*** (0.2960)	0.3943 (0.3135)	0.5401* (0.3802)	0.4657 (0.4085)	1.4303** (0.6102)
Elite U.S. Liberal Arts	1.1173*** (0.3813)	0.6279* (0.3754)	1.0698** (0.5190)	-0.6088 (1.4106)	2.3151** (1.0842)
Elite Foreign University	1.0061* (0.6050)	0.4759 (0.8224)	0.7595* (0.7938)	1.7158** (0.8587)	1.6165* (1.9880)
Other Foreign University	0.5442 (0.5785)	0.5777 (0.7688)	1.3733** (0.6902)	0.4188 (0.7157)	-1.0737 (1.3016)
Graduate Degree	0.0467 (0.2238)	0.0733 (0.2583)	0.3128 (0.3280)	0.4849 (0.4896)	0.4892 (0.6390)
Work Experience	0.3932** (0.1889)	0.4027 (0.2093)	0.4972 (0.3265)	0.1495 (0.3778)	1.8201** (0.8564)
#Professor Recommenders	0.0346 (0.1317)	-0.0178 (0.1339)	0.5511** (0.2554)	0.4701* (0.2652)	-0.0981 (0.2660)
#Prolific Recommenders	0.2635*** (0.0854)	0.3025** (0.1262)	0.1880 (0.1576)	0.3284** (0.1544)	0.5001* (0.3145)
#Math Courses	0.1689*** (0.0596)				
Advanced Math	0.2319 (0.1790)				
Missing GRE	0.1899 (0.7646)	-0.3259 (0.3100)	0.6282 (0.4275)	-0.0831 (0.3761)	-1.1514** (0.5702)
Missing Undergraduate GPA	-0.4745 (0.4891)	-0.4017 (0.5440)	-1.5492** (0.6893)	-1.1087* (0.6401)	0.4890 (1.1831)
Missing TOEFL/IELTS	-0.3786 (0.2432)	0.6188 (0.5657)	0.6745 (0.5190)	-0.0440 (0.5099)	-1.7224** (0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2677	0.1525	0.1982	0.1871	0.2475

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative** ($p < .1$): Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression AMEs

Table 15b: Graduate School Logistic Regression AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303** (0.0153)	-0.0164 (0.0100)	-0.0431** (0.0261)	-0.0298 (0.0224)	-0.0075 (0.0238)	-0.0218*** (0.0084)
Winsorized Age	0.0017** (0.0027)	0.0017** (0.0010)	0.0028 (0.0025)	0.0012 (0.0020)	-0.0022 (0.0038)	0.0016* (0.0009)
Female	0.0312*** (0.0129)	0.0254*** (0.0088)	0.0400*** (0.0220)	0.0071 (0.0187)	-0.0216 (0.0239)	0.0201*** (0.0075)
U.S. African/Hispanic/Native	0.0323 (0.0402)	0.0381*** (0.0137)	0.0869*** (0.0332)	0.0007 (0.0455)	-0.0352 (0.0487)	0.0384*** (0.0119)
U.S. Asian	-0.0413 (0.0322)	-0.0215 (0.0247)	-0.0244 (0.0936)	0.0468 (0.0391)	0.0499 (0.0513)	0.0062 (0.0185)
International Asian	-0.0646*** (0.0224)	-0.0070 (0.0157)	0.0704 (0.0543)	0.0265 (0.0363)	-0.0334 (0.0393)	0.0069 (0.0132)
International Other	0.0385 (0.0423)	0.0143 (0.0157)	0.1013*** (0.0385)	-0.0067 (0.0406)	0.0266 (0.0471)	0.0238* (0.0132)
GRE Verbal Percentile	0.0012*** (0.0005)	0.0011* (0.0006)	0.0056** (0.0025)	-0.0005 (0.0008)	0.0026*** (0.0010)	0.0015*** (0.0005)
GRE Quantitative Percentile	0.0118*** (0.0014)	0.0066* (0.0003)	0.0016* (0.0009)	0.0011 (0.0008)	-0.0006 (0.0007)	0.0006*** (0.0003)
GRE Analytic Percentile	0.0014*** (0.0003)	0.0008** (0.0004)	0.0002 (0.0010)	-0.0009 (0.0007)	-0.0009 (0.0009)	0.0007** (0.0003)
Undergraduate GPA	0.1699*** (0.0551)	0.0264 (0.0308)	0.1022** (0.0423)	0.0984 (0.0522)	0.0757* (0.0418)	0.0486*** (0.0176)
TOEFL/IELTS	0.0207 (0.0219)	0.0044 (0.0179)	0.0523* (0.0417)	0.0523* (0.0301)	-0.0198 (0.0259)	0.0326*** (0.0123)
Elite U.S. University	0.0877*** (0.0225)	0.0149 (0.0119)	0.0410 (0.0281)	0.0461*** (0.0264)	-0.0233*** (0.0271)	0.0409** (0.0098)
Elite U.S. Liberal Arts College	0.0859*** (0.0288)	0.0238* (0.0143)	0.0801** (0.0402)	-0.0130 (0.0761)	0.0667** (0.0480)	0.0420*** (0.0136)
Elite Foreign University	0.0463 (0.0463)	0.0311 (0.0311)	0.0591 (0.0591)	0.0481 (0.0481)	0.0850 (0.0850)	0.0241 (0.0241)
Other Foreign University	0.0416 (0.0441)	0.0219 (0.0291)	0.1028* (0.0527)	0.0227 (0.0391)	-0.0458 (0.0556)	0.0273 (0.0221)
Graduate Degree	0.0026 (0.0171)	0.0028 (0.0098)	0.0234 (0.0247)	0.0219 (0.0262)	0.0208 (0.0273)	0.0106 (0.0089)
Work Experience	0.0301** (0.0144)	0.0152 (0.0113)	0.0372 (0.0248)	0.0081 (0.0205)	0.0776** (0.0368)	0.0221** (0.0089)
#Professor Recommenders	0.0005 (0.0101)	-0.0007 (0.0051)	0.0209* (0.0190)	0.0209* (0.0142)	-0.0042 (0.0113)	0.0063 (0.0047)
#Prolific Recommenders	0.0202*** (0.0065)	0.0114** (0.0048)	0.0141 (0.0118)	0.0178** (0.0082)	0.0251* (0.0130)	0.0133*** (0.0038)
#Math Courses	0.0129*** (0.0045)					
Advanced Math	0.0177 (0.0137)					
Missing GRE	0.0145 (0.0585)	-0.0123 (0.0118)	0.0470 (0.0319)	-0.0045 (0.0204)	-0.0401* (0.0252)	-0.0062 (0.0088)
Missing Undergraduate GPA	-0.0363 (0.0375)	-0.0186 (0.0266)	-0.1160** (0.0524)	-0.0601* (0.0355)	0.0200 (0.0605)	-0.0291 (0.0179)
Missing TOEFL/IELTS	-0.0290 (0.0188)	0.0234 (0.0225)	0.0505 (0.0388)	-0.0034 (0.0276)	-0.0734* (0.0417)	0.0108 (0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	115.8637	23	0.0000			

Robust standard errors in parentheses. *p<0.10, **p<0.05, ***p<0.01.

- Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

A4: Graduate School Logistic Regression Results (Synthetic)

Table 16a: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997 (0.1633)	-0.0223 (0.2215)	0.2028 (0.3078)	-0.4791 (0.3477)	-0.5946 (0.4520)
Winsorized Age	-0.0509** (0.0201)	-0.0198 (0.0174)	0.0576* (0.0307)	-0.0164 (0.0357)	0.1203*** (0.0400)
Female	0.1941 (0.1402)	0.0923 (0.2004)	0.1389 (0.2526)	0.2595 (0.3255)	0.2119 (0.4310)
U.S. African/Hispanic/Native	0.3053 (0.3724)	-0.1457 (0.4305)	-0.2379 (0.4670)	-1.6401 (1.0720)	0.5413 (0.5306)
U.S. Asian	0.5663 (0.3642)	0.6016 (0.4452)	-0.7804 (1.1675)	-0.4305 (0.7936)	0.7122 (0.8863)
International Asian	-0.1428 (0.2169)	0.1630 (0.2803)	0.5780 (0.3740)	-0.4065 (0.4275)	0.1868 (0.6866)
International Other	-0.1231 (0.2111)	0.1395 (0.3046)	0.1097 (0.3237)	-0.1602 (0.3832)	-1.0807 (0.7583)
GRE Verbal Percentile	0.0167*** (0.0040)	0.0247** (0.0117)	0.0164 (0.0127)	-0.0029 (0.0100)	0.0132 (0.0184)
GRE Quantitative Percentile	0.0233*** (0.0090)	-0.0027 (0.0051)	0.0026 (0.0088)	0.0304** (0.0137)	0.0097 (0.0090)
GRE Analytic Percentile	0.0042 (0.0031)	0.0004 (0.0050)	-0.0073 (0.0092)	-0.0042 (0.0101)	0.0069 (0.0103)
Undergraduate GPA	0.2292 (0.4745)	0.6327 (0.4085)	1.3552*** (0.5796)	-1.3371** (0.5584)	-0.1002 (0.6152)
TOEFL/IELTS	-0.4673** (0.2157)	0.5141 (0.4185)	-0.7648 (0.6814)	0.2446 (0.7498)	-0.7354 (0.8025)
Elite U.S. University	-0.4850* (0.2560)	-0.2243 (0.3258)	0.0572 (0.3421)	0.4764 (0.4788)	0.3941 (0.4782)
Elite U.S. Liberal Arts College	0.2429 (0.3049)	0.1750 (0.4042)	0.2358 (0.4964)	0.4860 (0.9631)	0.9652 (0.7261)
Elite Foreign University	-0.1087 (0.3015)	-0.1622 (0.7484)	-1.1029 (0.5668)	0.4801 (0.7841)	3.0516*** (0.8673)
Other Foreign University	-0.1087 (0.1979)	0.2043 (0.2732)	0.2721 (0.3391)	0.1731 (0.4202)	0.3023 (0.5926)
Graduate Degree	-0.0186 (0.1624)	-0.0186 (0.2439)	-0.1661 (0.2765)	-0.0698*** (0.3320)	-0.2769 (0.3814)
Work Experience	0.2525* (0.1441)	-0.0391 (0.2396)	-0.0736 (0.2629)	-0.2650 (0.3417)	-1.1936** (0.4605)
#Professor Recommenders	-0.1059 (0.0909)	0.0911 (0.1138)	-0.1838 (0.1473)	0.2161 (0.1817)	0.4924** (0.1956)
#Profic Recommenders	0.0039 (0.0636)	0.2189** (0.1108)	-0.1743 (0.1723)	0.0673 (0.1642)	0.5134** (0.2213)
#Math Courses	0.0396 (0.0421)				
Advanced Math	0.1961 (0.1416)				
Missing GRE	-0.1093 (0.4328)	-0.2029 (0.2195)	-0.0081 (0.2689)	-0.0433 (0.3503)	-0.6040 (0.4017)
Missing Undergraduate GPA	0.1531 (0.1540)	0.0895 (0.2293)	-0.2455 (0.3015)	0.6298** (0.3125)	-0.3574 (0.4342)
Missing TOEFL/IELTS	-0.2950* (0.1640)	0.4698 (0.3480)	0.5752 (0.5466)	0.1346 (0.4345)	-0.0781*** (0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2251	690	737	486
Pseudo-R ²	0.0524	0.0366	0.0753	0.1046	0.2230

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive** ($p < .1$): GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative** ($p < .1$): Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A4: Graduate School Logistic Regression AMEs (Synthetic)

Table 16b: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204 (0.0167)	-0.0014 (0.0099)	0.0192 (0.0296)	-0.0298 (0.0216)	-0.0345 (0.0282)	-0.0089 (0.0086)
Winsorized Age	-0.0052** (0.0021)	-0.0009 (0.0038)	0.0054** (0.0029)	-0.0010 (0.0022)	0.0070*** (0.0023)	0.0004 (0.0008)
Female	0.0198 (0.0143)	0.0041 (0.0089)	0.0162 (0.0239)	0.0123 (0.0204)	0.0099 (0.0251)	0.0099 (0.0078)
U.S. African/Hispanic/Native	0.0312 (0.0381)	-0.0065 (0.0192)	-0.0225 (0.0441)	-0.1021 (0.0673)	0.0314 (0.0396)	-0.0437 (0.0147)
U.S. Asian	0.0679 (0.0372)	0.0266 (0.0199)	-0.0750 (0.1105)	-0.0268 (0.0495)	0.0413 (0.0545)	0.0241 (0.0179)
International Asian	-0.0146 (0.0222)	0.0073 (0.0125)	0.0546 (0.0353)	-0.0309 (0.0263)	0.0108 (0.0399)	0.0075 (0.0112)
International Other	-0.0126 (0.0236)	0.0062 (0.0136)	0.0104 (0.0306)	-0.0100 (0.0237)	-0.0027 (0.0443)	-0.0014 (0.0106)
GRE Verbal Percentile	0.0017*** (0.0005)	0.0011** (0.0005)	0.0015 (0.0012)	-0.0002 (0.0006)	0.0008 (0.0010)	0.0008** (0.0004)
GRE Quantitative Percentile	0.0024*** (0.0009)	-0.0001 (0.0002)	0.0003 (0.0008)	0.0019*** (0.0009)	0.0006 (0.0006)	0.0003 (0.0002)
GRE Analytic Percentile	0.0004 (0.0003)	0.0000 (0.0003)	-0.0007 (0.0009)	-0.0003 (0.0006)	0.0004 (0.0006)	0.0000 (0.0002)
Undergraduate GPA	0.0234 (0.0485)	0.0282 (0.0223)	0.1281*** (0.0553)	0.0820** (0.0398)	0.0106 (0.0357)	0.0130 (0.0158)
TOEFL/IELTS	-0.0478** (0.0221)	0.0229 (0.0187)	-0.0723 (0.0645)	0.0152 (0.0466)	-0.0427 (0.0469)	0.0050 (0.0187)
Elite U.S. University	-0.0466* (0.0261)	-0.0100 (0.0145)	0.0207 (0.0324)	0.0297 (0.0297)	0.0229 (0.0274)	0.0036 (0.0108)
Elite U.S. Liberal Arts College	0.0248 (0.0312)	0.0078 (0.0180)	0.0223 (0.0469)	0.0303 (0.0602)	0.0502 (0.0410)	0.0217 (0.0160)
Elite Foreign University	-0.0166 (0.0308)	-0.0512 (0.0337)	0.0700 (0.0537)	0.0299 (0.0400)	0.1771*** (0.0484)	0.0156 (0.0179)
Other Foreign University	-0.0111 (0.0202)	0.0091 (0.0122)	0.0257 (0.0321)	0.0108 (0.0267)	0.0175 (0.0344)	0.0116 (0.0103)
Graduate Degree	-0.0020 (0.0166)	0.0079 (0.0109)	-0.0015 (0.0261)	-0.0642*** (0.0206)	-0.0161 (0.0220)	-0.0063 (0.0085)
Work Experience	0.0258* (0.0146)	-0.0017 (0.0107)	-0.0070 (0.0248)	-0.0165 (0.0213)	-0.0692** (0.0275)	-0.0093 (0.0086)
#Professor Recommenders	-0.0108 (0.0093)	0.0041 (0.0051)	-0.0173 (0.0140)	0.0136 (0.0114)	-0.0286*** (0.0110)	-0.0027 (0.0042)
#Profic Recommenders	0.0004 (0.0097)	0.0096* (0.0050)	-0.0165 (0.0164)	0.0042 (0.0102)	0.0298** (0.0126)	0.0033* (0.0045)
#Math Courses	0.0037 (0.0043)	0.0037 (0.0043)				
Advanced Math	0.0201 (0.0145)					
Missing GRE	-0.0204 (0.0443)	-0.0090 (0.0098)	-0.0001 (0.0254)	-0.0027 (0.0218)	-0.0350 (0.0239)	-0.0098 (0.0079)
Missing Undergraduate GPA	0.0157 (0.0158)	0.0040 (0.0102)	-0.0232 (0.0285)	0.0392* (0.0201)	-0.0207 (0.0254)	0.0019 (0.0083)
Missing TOEFL/IELTS	-0.0305* (0.0168)	0.0218 (0.0156)	0.0544 (0.0517)	0.0684 (0.0272)	-0.1265*** (0.0434)	0.0063 (0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

A5: Simulated Wald Test Results (Same)

Table 17a: Simulated Wald Test Results (Same)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848*** (1.1315)	-1.5822*** (0.0956)	-28.5524*** (1.4963)	-1.3677*** (0.0500)	-10.4586*** (0.7236)	-1.9105*** (0.1070)	-10.2610*** (0.4815)	-1.8514*** (0.0628)
GRE Quantitative Percentile	0.1509*** (0.0131)	0.0146*** (0.0012)	0.3245*** (0.0179)	0.0155*** (0.0007)	0.1058*** (0.0088)	0.0193*** (0.0014)	0.1154*** (0.0070)	0.0208*** (0.0011)
Gender	0.4714*** (0.1475)	0.0455*** (0.0140)	0.7880*** (0.1696)	0.0377*** (0.0081)	-0.1384 (0.1095)	-0.0253 (0.0200)	-0.0497 (0.0895)	-0.0090 (0.0161)
Demographic 1	-0.6287*** (0.2088)	-0.0607*** (0.0200)	-1.8488*** (0.2445)	-0.0886*** (0.0115)	-0.3486** (0.1467)	-0.0637** (0.0267)	-0.3079*** (0.1123)	-0.0556*** (0.0202)
Demographic 2	0.1531 (0.2072)	0.0148 (0.0200)	0.3703* (0.1929)	0.0177* (0.0091)	0.1756 (0.1558)	0.0321 (0.0284)	0.0282 (0.1069)	0.0051 (0.0193)
Advanced Math	0.2889* (0.1612)	0.0279* (0.0156)			0.3267*** (0.1112)	0.0597*** (0.0202)		
Committee Score	0.2241*** (0.0413)	0.0216*** (0.0039)	0.3994*** (0.0506)	0.0191*** (0.0023)	0.1026*** (0.0287)	0.0187*** (0.0052)	0.0757*** (0.0228)	0.0137*** (0.0041)
Non-Economics 1			0.8376*** (0.2297)	0.0401*** (0.0107)			1.9483*** (0.1236)	0.3515*** (0.0205)
Observations Pseudo-R ²	5000	0.3251			5000	0.1431		
Parameters LL AIC	14	-1130.4	2288.8		14	-2691.4	5410.8	
Wald Statistic DF P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Wald Test Results (Different)

Table 17b: Simulated Wald Test Results (Different)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291*** (1.1274)	-1.5784*** (0.0955)	-8.4753*** (0.8673)	-0.3642*** (0.0412)	-10.4581*** (0.7237)	-1.9104*** (0.1070)	-3.2236*** (0.4018)	-0.5997*** (0.0728)
GRE Quantitative Percentile	0.1502*** (0.0131)	0.0145*** (0.0012)	0.0402*** (0.0128)	0.0017*** (0.0006)	0.1058*** (0.0088)	0.0193*** (0.0014)	0.0034 (0.0062)	0.0006 (0.0012)
Gender	0.4702*** (0.1473)	0.0454*** (0.0140)	0.5942*** (0.1820)	0.0255*** (0.0079)	-0.1372 (0.1095)	-0.0251 (0.0200)	0.0375 (0.0868)	0.0070 (0.0161)
Demographic 1	-0.6246*** (0.2085)	-0.0604*** (0.0200)	-0.2541 (0.2393)	-0.0109 (0.0103)	-0.3544** (0.1467)	-0.0647** (0.0267)	-0.1152 (0.1103)	-0.0214 (0.0205)
Demographic 2	0.1555 (0.2070)	0.0150 (0.0200)	0.0081 (0.2109)	0.0003 (0.0091)	0.1713 (0.1557)	0.0313 (0.0284)	-0.0562 (0.1030)	-0.0105 (0.0192)
Advanced Math	0.2886* (0.1611)	0.0279* (0.0156)			0.3284*** (0.1112)	0.0600*** (0.0202)		
Committee Score	0.2241*** (0.0413)	0.0217*** (0.0039)	0.3817*** (0.0528)	0.0164*** (0.0025)	0.1026*** (0.0287)	0.0187*** (0.0052)	0.1808*** (0.0228)	0.0336*** (0.0041)
Non-Economics 1			-0.0452 (0.1995)	-0.0019 (0.0086)			1.1393*** (0.1219)	0.2120*** (0.0219)
Observations Pseudo-R ²	5000	0.1488			5000	0.0758		
Parameters LL AIC	14	-1155.7	2339.5		14	-2747.2	5522.4	
Wald Statistic DF P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Unrestricted Dynamic Estimates (Optimal)

Table 18a: Simulated Unrestricted Dynamic Estimates (Optimal)

	Advanced Math		Committee Score		Transition		Accept		Success	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027** (0.7564)	-0.4663** (0.1830)	4.8637*** (1.3238)		-11.5885*** (1.0232)	-2.2558*** (0.1454)	-13.1577*** (2.0081)	-2.3739*** (0.2886)	-9.4807*** (1.3435)	-1.3248*** (0.1562)
GRE Quantitative Percentile	0.0208** (0.0090)	0.0051** (0.0022)	0.0008 (0.0150)		0.1236*** (0.0118)	0.0241*** (0.0018)	0.1372*** (0.0223)	0.0248*** (0.0033)	0.1150*** (0.0145)	0.0161*** (0.0016)
Gender	-0.2335* (0.1309)	-0.0572* (0.0319)	-0.0783 (0.2113)		0.2266 (0.1464)	0.0441 (0.0283)	0.1289 (0.2502)	0.0233 (0.0451)	0.2078 (0.1766)	0.0290 (0.0246)
Demographic 1	0.3691** (0.1732)	0.0905** (0.0421)	-0.3430 (0.2885)		-0.3014 (0.1988)	-0.0587 (0.0386)	-0.2640 (0.3744)	-0.0476 (0.0672)	-0.3380 (0.2456)	-0.0472 (0.0342)
Demographic 2	0.0925 (0.1911)	0.0227 (0.0468)	-0.1806 (0.3001)		0.1538 (0.2215)	0.0299 (0.0431)	0.1708 (0.3844)	0.0308 (0.0693)	-0.1354 (0.2652)	-0.0189 (0.0371)
Advanced Math			0.6870*** (0.2024)				0.7737*** (0.2535)	0.1396*** (0.0443)	0.6827*** (0.1770)	0.0954*** (0.0237)
Committee Score							0.0783 (0.0602)	0.0141 (0.0108)	0.0467 (0.0672)	0.0065 (0.0094)
Missing Committee Score									0.2177 (0.1896)	0.0304 (0.0264)
Year Transition More	0.0664 (0.1281)	0.0163 (0.0314)	0.0932 (0.2069)		0.9840*** (0.1472)	0.1915*** (0.0263)	-0.9251*** (0.2513)	-0.1669*** (0.0427)	-0.0466 (0.1740)	-0.0065 (0.0243)
Program Rank									-0.0309*** (0.0019)	-0.0043*** (0.0002)
$\sqrt{\text{MSE}}$			1.9465*** (0.0673)							
Obs Param LL AIC	1000	38	-2655.7		5387.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Unrestricted Dynamic Estimates (Suboptimal)

Table 18b: Simulated Unrestricted Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score		Transition		Accept		Success	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027** (0.7564)	-0.4663** (0.1830)	5.2717*** (1.2701)		-11.2295*** (1.0144)	-2.1801*** (0.1474)	-6.7296*** (1.7376)	-1.3296*** (0.3202)	-9.7587*** (1.3345)	-1.3686*** (0.1538)
GRE Quantitative Percentile	0.0208** (0.0090)	0.0051** (0.0022)	-0.0016 (0.0144)		0.1226*** (0.0117)	0.0238*** (0.0018)	0.0550*** (0.0191)	0.0109*** (0.0036)	0.1175*** (0.0145)	0.0165*** (0.0016)
Gender	-0.2335* (0.1309)	-0.0572* (0.0319)	-0.1023 (0.2051)		0.2575* (0.1469)	0.0500* (0.0283)	0.0098 (0.2368)	0.0019 (0.0468)	0.1984 (0.1766)	0.0278 (0.0247)
Demographic 1	0.3691** (0.1732)	0.0905** (0.0421)	-0.4893* (0.2701)		-0.8293*** (0.1966)	-0.1610*** (0.0371)	-0.3340 (0.3394)	-0.0660 (0.0669)	-0.3897 (0.2462)	-0.0547 (0.0343)
Demographic 2	0.0925 (0.1911)	0.0227 (0.0468)	-0.2744 (0.2740)		0.0797 (0.2135)	0.0155 (0.0414)	0.1764 (0.3346)	0.0348 (0.0659)	-0.2068 (0.2636)	-0.0290 (0.0370)
Advanced Math			0.7219*** (0.1966)				0.3580 (0.2375)	0.0707 (0.0465)	0.6387*** (0.1763)	0.0896*** (0.0238)
Committee Score							0.2962*** (0.0592)	0.0585*** (0.0106)	0.0626 (0.0657)	0.0088 (0.0092)
Missing Committee Score									0.2535 (0.1896)	0.0356 (0.0264)
Year Transition More	0.0664 (0.1281)	0.0163 (0.0314)	0.0031 (0.2010)		1.1108*** (0.1479)	0.2156*** (0.0257)	-0.7185*** (0.2397)	-0.1420*** (0.0455)	-0.0473 (0.1757)	-0.0066 (0.0247)
Program Rank									-0.0303*** (0.0019)	-0.0043*** (0.0002)
$\sqrt{MSE}$			1.9186*** (0.0682)							
Obs Param LL AIC	1000	38	-2696.1		5468.2					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Restricted Dynamic Estimates (Optimal)

Table 19a: Simulated Restricted Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-2.1626*** (0.7577)	4.6976*** (1.3258)	-0.3078*** (0.0655)	0.7351 (0.6465, 0.8358)	0.0503 (0.1841)	51.26% (42.30%, 60.13%)	-9.5097*** (1.2612)	
GRE Quantitative Percentile	0.0240*** (0.0090)	0.0028 (0.0150)					0.1148*** (0.0137)	
Gender	-0.2129 (0.1303)	-0.0649 (0.2112)					0.2357** (0.1084)	
Demographic 1	0.3565** (0.1724)	-0.3526 (0.2903)					-0.3177** (0.1549)	
Demographic 2	0.0948 (0.1904)	-0.1805 (0.3010)					0.0020 (0.1594)	
Advanced Math		0.6919*** (0.2026)					0.6157*** (0.1367)	
Committee Score							0.0496 (0.0383)	
Missing Committee Score							0.2183 (0.1819)	
Year Transition More	0.0743 (0.1277)	0.0967 (0.2069)	-0.0466 (0.0675)	0.7017 (0.6137, 0.8022)	0.5233** (0.2398)	63.96% (54.65%, 72.33%)	-0.0124 (0.1720)	
Program Rank							-0.0310*** (0.0019)	
$\sqrt{\text{MSE}}$		1.9468*** (0.0673)						
Sigma								0.1390*** (0.0189)
Obs Param LL AIC	1000	29	-2683.0	5423.9				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A5: Simulated Restricted Dynamic Estimates (Suboptimal)

Table 19b: Simulated Restricted Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-2.1764*** (0.7589)	4.5383*** (1.2591)	-0.2386*** (0.0803)	0.7878 (0.6730, 0.9220)	0.0226 (0.2017)	50.57% (40.79%, 60.30%)	-9.4138*** (1.4009)	
GRE Quantitative Percentile	0.0242*** (0.0090)	0.0072 (0.0143)					0.1066*** (0.0150)	
Gender	-0.2179* (0.1306)	-0.0641 (0.2029)					0.2086* (0.1206)	
Demographic 1	0.3420** (0.1728)	-0.5643** (0.2716)					-0.5037*** (0.1798)	
Demographic 2	0.1011 (0.1906)	-0.2640 (0.2738)					-0.0204 (0.1843)	
Advanced Math		0.7292*** (0.1970)					0.5164*** (0.1483)	
Committee Score							0.1934*** (0.0382)	
Missing Committee Score							0.2727 (0.1887)	
Year Transition More	0.0706 (0.1278)	0.0108 (0.2007)	-0.1333* (0.0712)	0.6895 (0.5886, 0.8075)	0.4714* (0.2663)	62.11% (51.59%, 71.59%)	-0.0673 (0.1760)	
Program Rank							-0.0305*** (0.0019)	
$\sqrt{\text{MSE}}$		1.9197*** (0.0684)						
Sigma								0.1776*** (0.0305)
Obs Param LL AIC	1000	29	-2742.2	5542.4				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

A5: Simulated Unrestricted Static Estimates (Optimal)

Table 20a: Simulated Unrestricted Static Estimates (Optimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.5885*** (1.0232)	-2.2558*** (0.1454)	-13.1577*** (2.0081)	-2.3739*** (0.2886)	-8.1374*** (1.1381)	-1.1678*** (0.1344)	-9.4807*** (1.3435)	-1.3248*** (0.1562)
GRE Quantitative Percentile	0.1236*** (0.0118)	0.0241*** (0.0018)	0.1372*** (0.0223)	0.0248*** (0.0033)	0.1079*** (0.0133)	0.0155*** (0.0015)	0.1150*** (0.0145)	0.0161*** (0.0016)
Gender	0.2266 (0.1464)	0.0441 (0.0283)	0.1289 (0.2502)	0.0233 (0.0451)	0.1399 (0.1739)	0.0201 (0.0249)	0.2078 (0.1766)	0.0290 (0.0246)
Demographic 1	-0.3014 (0.1988)	-0.0587 (0.0386)	-0.2640 (0.3744)	-0.0476 (0.0672)	-0.2533 (0.2359)	-0.0364 (0.0338)	-0.3380 (0.2456)	-0.0472 (0.0342)
Demographic 2	0.1538 (0.2215)	0.0299 (0.0431)	0.1708 (0.3844)	0.0308 (0.0693)	-0.1179 (0.2581)	-0.0169 (0.0371)	-0.1354 (0.2652)	-0.0189 (0.0371)
Advanced Math			0.7737*** (0.2535)	0.1396*** (0.0443)			0.6827*** (0.1770)	0.0954*** (0.0237)
Committee Score			0.0783 (0.0602)	0.0141 (0.0108)			0.0467 (0.0672)	0.0065 (0.0094)
Missing Committee Score							0.2177 (0.1896)	0.0304 (0.0264)
Year Transition More	0.9840*** (0.1472)	0.1915*** (0.0263)	-0.9251*** (0.2513)	-0.1669*** (0.0427)	-0.0587 (0.1690)	-0.0084 (0.0242)	-0.0466 (0.1740)	-0.0065 (0.0243)
Program Rank					-0.0310*** (0.0019)	-0.0045*** (0.0002)	-0.0309*** (0.0019)	-0.0043*** (0.0002)
Obs Param LL AIC	1000	31	-1622.9	3307.8				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A5: Simulated Unrestricted Static Estimates (Suboptimal)

Table 20b: Simulated Unrestricted Static Estimates (Suboptimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.2295*** (1.0144)	-2.1801*** (0.1474)	-6.7296*** (1.7376)	-1.3296*** (0.3202)	-8.2825*** (1.1392)	-1.1913*** (0.1341)	-9.7587*** (1.3345)	-1.3686*** (0.1538)
GRE Quantitative Percentile	0.1226*** (0.0117)	0.0238*** (0.0018)	0.0550*** (0.0191)	0.0109*** (0.0036)	0.1097*** (0.0133)	0.0158*** (0.0015)	0.1175*** (0.0145)	0.0165*** (0.0016)
Gender	0.2575* (0.1469)	0.0500* (0.0283)	0.0098 (0.2368)	0.0019 (0.0468)	0.1355 (0.1741)	0.0195 (0.0250)	0.1984 (0.1766)	0.0278 (0.0247)
Demographic 1	-0.8293*** (0.1966)	-0.1610*** (0.0371)	-0.3340 (0.3394)	-0.0660 (0.0669)	-0.2776 (0.2344)	-0.0399 (0.0337)	-0.3897 (0.2462)	-0.0547 (0.0343)
Demographic 2	0.0797 (0.2135)	0.0155 (0.0414)	0.1764 (0.3346)	0.0348 (0.0659)	-0.1809 (0.2566)	-0.0260 (0.0369)	-0.2068 (0.2636)	-0.0290 (0.0370)
Advanced Math			0.3580 (0.2375)	0.0707 (0.0465)			0.6387*** (0.1763)	0.0896*** (0.0238)
Committee Score			0.2962*** (0.0592)	0.0585*** (0.0106)			0.0626 (0.0657)	0.0088 (0.0092)
Missing Committee Score							0.2535 (0.1896)	0.0356 (0.0264)
Year Transition More	1.1108*** (0.1479)	0.2156*** (0.0257)	-0.7185*** (0.2397)	-0.1420*** (0.0455)	-0.0767 (0.1688)	-0.0110 (0.0243)	-0.0473 (0.1757)	-0.0066 (0.0247)
Program Rank					-0.0304*** (0.0019)	-0.0044*** (0.0002)	-0.0303*** (0.0019)	-0.0043*** (0.0002)
Obs Param LL AIC	1000	31	-1647.5	3357.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Restricted Static Estimates (Optimal)

Table 21a: Simulated Restricted Static Estimates (Optimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.6911*** (0.1817)	66.62% (58.30%, 74.02%)	0.2080 (0.2148)	55.18% (44.69%, 65.23%)	-8.0508*** (1.1629)	-9.5892*** (1.3344)	
GRE Quantitative Percentile					0.1061*** (0.0134)	0.1155*** (0.0146)	
Gender					0.1997** (0.1017)	0.1694 (0.1361)	
Demographic 1					-0.2479* (0.1351)	-0.3121 (0.1916)	
Demographic 2					0.0408 (0.1509)	-0.0331 (0.1966)	
Advanced Math						0.6628*** (0.1426)	
Committee Score						0.0557 (0.0426)	
Missing Committee Score						0.2170 (0.1838)	
Year Transition More	-0.8478*** (0.2178)	46.09% (39.63%, 52.68%)	0.7486** (0.3087)	72.24% (60.18%, 81.76%)	-0.0606 (0.1667)	-0.0276 (0.1730)	
Program Rank					-0.0310*** (0.0019)	-0.0310*** (0.0019)	
Sigma							0.1922*** (0.0246)
Obs Param LL AIC	1000	22	-1623.7	3291.3			

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A5: Simulated Restricted Static Estimates (Suboptimal)

Table 21b: Simulated Restricted Static Estimates (Suboptimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.8247*** (0.2158)	69.52% (59.91%, 77.69%)	0.2858 (0.2418)	57.10% (45.31%, 68.13%)	-8.5784*** (1.1959)	-8.3580*** (1.4863)	
GRE Quantitative Percentile					0.1141*** (0.0138)	0.0950*** (0.0158)	
Gender					0.2291** (0.1093)	0.1303 (0.1417)	
Demographic 1					-0.5672*** (0.1465)	-0.3661* (0.2059)	
Demographic 2					-0.0218 (0.1599)	-0.0705 (0.2104)	
Advanced Math						0.5399*** (0.1477)	
Committee Score						0.1918*** (0.0412)	
Missing Committee Score						0.1703 (0.1949)	
Year Transition More	-1.0834*** (0.2489)	43.57% (37.09%, 50.27%)	0.5451* (0.3249)	69.66% (57.04%, 79.87%)	-0.0713 (0.1664)	-0.1035 (0.1784)	
Program Rank					-0.0306*** (0.0019)	-0.0305*** (0.0019)	
Sigma							0.2211*** (0.0320)
Obs Param LL AIC	1000	22	-1662.3	3368.6			

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A5: Simulated Unrestricted Myopic Estimates (Optimal)

Table 22a: Simulated Unrestricted Myopic Estimates (Optimal)

	Advanced Math		Committee Score		Transition		Accept		Success	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000 (0.0894)	0.0000 (0.0224)	4.6668*** (0.1940)		-11.5885*** (1.0232)	-2.2558*** (0.1454)	-13.1577*** (2.0081)	-2.3739*** (0.2886)	-9.4807*** (1.3435)	-1.3248*** (0.1562)
GRE Quantitative Percentile					0.1236*** (0.0118)	0.0241*** (0.0018)	0.1372*** (0.0223)	0.0248*** (0.0033)	0.1150*** (0.0145)	0.0161*** (0.0016)
Gender					0.2266 (0.1464)	0.0441 (0.0283)	0.1289 (0.2502)	0.0233 (0.0451)	0.2078 (0.1766)	0.0290 (0.0246)
Demographic 1					-0.3014 (0.1988)	-0.0587 (0.0386)	-0.2640 (0.3744)	-0.0476 (0.0672)	-0.3380 (0.2456)	-0.0472 (0.0342)
Demographic 2					0.1538 (0.2215)	0.0299 (0.0431)	0.1708 (0.3844)	0.0308 (0.0693)	-0.1354 (0.2652)	-0.0189 (0.0371)
Advanced Math			0.6763*** (0.2006)				0.7737*** (0.2535)	0.1396*** (0.0443)	0.6827*** (0.1770)	0.0954*** (0.0237)
Committee Score							0.0783 (0.0602)	0.0141 (0.0108)	0.0467 (0.0672)	0.0065 (0.0094)
Missing Committee Score									0.2177 (0.1896)	0.0304 (0.0264)
Year Transition More	0.0480 (0.1265)	0.0120 (0.0316)	0.1023 (0.2069)		0.9840*** (0.1472)	0.1915*** (0.0263)	-0.9251*** (0.2513)	-0.1669*** (0.0427)	-0.0466 (0.1740)	-0.0065 (0.0243)
Program Rank									-0.0309*** (0.0019)	-0.0043*** (0.0002)
$\sqrt{MSE}$			1.9512*** (0.0674)							
Obs Param LL AIC	1000	30	-2666.4		5392.8					

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A5: Simulated Unrestricted Myopic Estimates (Suboptimal)

Table 22b: Simulated Unrestricted Myopic Estimates (Suboptimal)

	Advanced Math		Committee Score Coefficient	Transition		Accept		Success	
	Coefficient	AME		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000 (0.0894)	0.0000 (0.0224)	4.7588*** (0.1881)	-11.2295*** (1.0144)	-2.1801*** (0.1474)	-6.7296*** (1.7376)	-1.3296*** (0.3202)	-9.7587*** (1.3345)	-1.3686*** (0.1538)
GRE Quantitative Percentile				0.1226*** (0.0117)	0.0238*** (0.0018)	0.0550*** (0.0191)	0.0109*** (0.0036)	0.1175*** (0.0145)	0.0165*** (0.0016)
Gender				0.2575* (0.1469)	0.0500* (0.0283)	0.0098 (0.2368)	0.0019 (0.0468)	0.1984 (0.1766)	0.0278 (0.0247)
Demographic 1				-0.8293*** (0.1966)	-0.1610*** (0.0371)	-0.3340 (0.3394)	-0.0660 (0.0669)	-0.3897 (0.2462)	-0.0547 (0.0343)
Demographic 2				0.0797 (0.2135)	0.0155 (0.0414)	0.1764 (0.3346)	0.0348 (0.0659)	-0.2068 (0.2636)	-0.0290 (0.0370)
Advanced Math			0.7040*** (0.1957)			0.3580 (0.2375)	0.0707 (0.0465)	0.6387*** (0.1763)	0.0896*** (0.0238)
Committee Score						0.2962*** (0.0592)	0.0585*** (0.0106)	0.0626 (0.0657)	0.0088 (0.0092)
Missing Committee Score								0.2535 (0.1896)	0.0356 (0.0264)
Year Transition More	0.0480 (0.1265)	0.0120 (0.0316)	0.0218 (0.2022)	1.1108*** (0.1479)	0.2156*** (0.0257)	-0.7185*** (0.2397)	-0.1420*** (0.0455)	-0.0473 (0.1757)	-0.0066 (0.0247)
Program Rank								-0.0303*** (0.0019)	-0.0043*** (0.0002)
$\sqrt{\text{MSE}}$			1.9289*** (0.0684)						
Obs Param LL AIC	1000	30	-2708.0	5475.9					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Restricted Myopic Estimates (Optimal)

Table 23a: Simulated Restricted Myopic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0146 (0.0889)	4.6803*** (0.1930)	-0.3118*** (0.0659)	0.7321 (0.6433, 0.8331)	0.0452 (0.1841)	51.13% (42.17%, 60.01%)	-9.5732*** (1.2633)	
GRE Quantitative Percentile							0.1156*** (0.0138)	
Gender							0.2266** (0.1080)	
Demographic 1							-0.3082** (0.1533)	
Demographic 2							0.0055 (0.1579)	
Advanced Math		0.6789*** (0.2009)					0.5999*** (0.1366)	
Committee Score							0.0496 (0.0382)	
Missing Committee Score							0.2197 (0.1818)	
Year Transition More	0.0542 (0.1261)	0.1004 (0.2067)	-0.0430 (0.0679)	0.7013 (0.6129, 0.8024)	0.5323** (0.2407)	64.05% (54.67%, 72.46%)	-0.0098 (0.1720)	
Program Rank							-0.0310*** (0.0019)	
$\sqrt{\text{MSE}}$		1.9514*** (0.0674)						
Sigma								0.1387*** (0.0190)
Obs Param LL AIC	1000	21	-2694.3	5430.7				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A5: Simulated Restricted Myopic Estimates (Suboptimal)

Table 23b: Simulated Restricted Myopic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0122 (0.0891)	4.8050*** (0.1878)	-0.2369*** (0.0811)	0.7891 (0.6731, 0.9250)	0.0217 (0.2029)	50.54% (40.71%, 60.33%)	-9.5133*** (1.3931)	
GRE Quantitative Percentile							0.1081*** (0.0149)	
Gender							0.2007* (0.1204)	
Demographic 1							-0.5140*** (0.1790)	
Demographic 2							-0.0254 (0.1847)	
Advanced Math		0.7027*** (0.1961)					0.5083*** (0.1484)	
Committee Score							0.1894*** (0.0385)	
Missing Committee Score							0.2765 (0.1884)	
Year Transition More	0.0511 (0.1262)	0.0095 (0.2016)	-0.1323* (0.0716)	0.6913 (0.5895, 0.8107)	0.4816* (0.2684)	62.32% (51.71%, 71.88%)	-0.0638 (0.1759)	
Program Rank							-0.0305*** (0.0019)	
$\sqrt{\text{MSE}}$		1.9289*** (0.0684)						
Sigma								0.1794*** (0.0308)
Obs Param LL AIC	1000	21	-2754.8	5551.6				

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A5: Simulated Unrestricted Mat-Dynamic Estimates (Optimal)

Table 24a: Simulated Unrestricted Matriculation-Dynamic Estimates (Optimal)

	Advanced Math		Committee Score		Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	4.8637***		-11.5885***	-2.2558***	-13.1577***	-2.3739***	-9.4807***	-1.3248***	-4.6066	-0.6127
	(0.7564)	(0.1830)	(1.3238)		(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.3435)	(0.1562)	(4.6480)	(0.6187)
GRE Quantitative Percentile	0.0208**	0.0051**	0.0008		0.1236***	0.0241***	0.1372***	0.0248***	0.1150***	0.0161***	0.0593	0.0079
	(0.0090)	(0.0022)	(0.0150)		(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0145)	(0.0016)	(0.0459)	(0.0061)
Gender	-0.2335*	-0.0572*	-0.0783		0.2266	0.0441	0.1289	0.0233	0.2078	0.0290	0.4885	0.0650
	(0.1309)	(0.0319)	(0.2113)		(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1766)	(0.0246)	(0.5063)	(0.0657)
Demographic 1	0.3691**	0.0905**	-0.3430		-0.3014	-0.0587	-0.2640	-0.0476	-0.3380	-0.0472	0.3587	0.0477
	(0.1732)	(0.0421)	(0.2885)		(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2456)	(0.0342)	(0.7459)	(0.1001)
Demographic 2	0.0925	0.0227	-0.1806		0.1538	0.0299	0.1708	0.0308	-0.1354	-0.0189	0.1834	0.0244
	(0.1911)	(0.0468)	(0.3001)		(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2652)	(0.0371)	(0.7962)	(0.1069)
Advanced Math			0.6870***				0.7737***	0.1396***	0.6827***	0.0954***	0.8106	0.1078
			(0.2024)				(0.2535)	(0.0443)	(0.1770)	(0.0237)	(0.5773)	(0.0733)
Committee Score							0.0783	0.0141	0.0467	0.0065	-0.7402***	-0.0985***
							(0.0602)	(0.0108)	(0.0672)	(0.0094)	(0.2064)	(0.0235)
Missing Committee Score									0.2177	0.0304		
									(0.1896)	(0.0264)		
Year Transition/Matriculate More	0.0664	0.0163	0.0932		0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0466	-0.0065	3.3280***	0.4427***
	(0.1281)	(0.0314)	(0.2069)		(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1740)	(0.0243)	(0.5674)	(0.0331)
Program Rank									-0.0309***	-0.0043***		
									(0.0019)	(0.0002)		
$\sqrt{\text{MSE}}$			1.9465***									
			(0.0673)									
Obs Param LL AIC	1000	46	-2708.1		5508.3							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Unrestricted Mat-Dynamic Estimates (Suboptimal)

Table 24b: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score		Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027** (0.7564)	-0.4663** (0.1830)	5.2717*** (1.2701)		-11.2295*** (1.0144)	-2.1801*** (0.1474)	-6.7296*** (1.7376)	-1.3296*** (0.3202)	-9.7587*** (1.3345)	-1.3686*** (0.1538)	-1.0400 (3.6083)	-0.1351 (0.4721)
GRE Quantitative Percentile	0.0208** (0.0090)	0.0051** (0.0022)	-0.0016 (0.0144)		0.1226*** (0.0117)	0.0238*** (0.0018)	0.0550*** (0.0191)	0.0109*** (0.0036)	0.1175*** (0.0145)	0.0165*** (0.0016)	0.0129 (0.0391)	0.0017 (0.0051)
Gender	-0.2335* (0.1309)	-0.0572* (0.0319)	-0.1023 (0.2051)		0.2575* (0.1469)	0.0500* (0.0283)	0.0098 (0.2368)	0.0019 (0.0468)	0.1984 (0.1766)	0.0278 (0.0247)	0.1298 (0.4717)	0.0169 (0.0608)
Demographic 1	0.3691** (0.1732)	0.0905** (0.0421)	-0.4893* (0.2701)		-0.8293*** (0.1966)	-0.1610*** (0.0371)	-0.3340 (0.3394)	-0.0660 (0.0669)	-0.3897 (0.2462)	-0.0547 (0.0343)	0.6327 (0.6281)	0.0822 (0.0823)
Demographic 2	0.0925 (0.1911)	0.0227 (0.0468)	-0.2744 (0.2740)		0.0797 (0.2135)	0.0155 (0.0414)	0.1764 (0.3346)	0.0348 (0.0659)	-0.2068 (0.2636)	-0.0290 (0.0370)	-0.2222 (0.6864)	-0.0289 (0.0880)
Advanced Math			0.7219*** (0.1966)				0.3580 (0.2375)	0.0707 (0.0465)	0.6387*** (0.1763)	0.0896*** (0.0238)	0.5877 (0.5321)	0.0763 (0.0674)
Committee Score							0.2962*** (0.0592)	0.0585*** (0.0106)	0.0626 (0.0657)	0.0088 (0.0092)	-0.6092*** (0.1637)	-0.0791*** (0.0176)
Missing Committee Score									0.2535 (0.1896)	0.0356 (0.0264)		
Year Transition/Matriculate More	0.0664 (0.1281)	0.0163 (0.0314)	0.0031 (0.2010)		1.1108*** (0.1479)	0.2156*** (0.0257)	-0.7185*** (0.2397)	-0.1420*** (0.0455)	-0.0473 (0.1757)	-0.0066 (0.0247)	3.4846*** (0.6046)	0.4526*** (0.0349)
Program Rank									-0.0303*** (0.0019)	-0.0043*** (0.0002)		
$\sqrt{\text{MSE}}$			1.9186*** (0.0682)									
Obs Param LL AIC	1000	46	-2749.9		5591.9							

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A5: Simulated Restricted Mat-Dynamic Estimates (Optimal)

Table 25a: Simulated Restricted Matriculation-Dynamic Estimates (Optimal)

Intercept	-2.1963*** (0.7585)	5.4976*** (1.3429)	-1.8249*** (0.1759)	0.1612 (0.1142, 0.2276)	0.0339 (0.2008)	50.85% (41.11%, 60.53%)	-9.4675*** (1.2874)	-9.8786** (4.0907)
GRE Quantitative Percentile	0.0244*** (0.0090)	-0.0072 (0.0153)					0.1153*** (0.0140)	0.1186*** (0.0397)
Gender	-0.2210* (0.1307)	-0.0965 (0.2096)					0.1845 (0.1213)	0.7496* (0.4548)
Demographic 1	0.3598** (0.1729)	-0.3130 (0.2878)					-0.3019* (0.1729)	0.1162 (0.7211)
Demographic 2	0.0946 (0.1907)	-0.1832 (0.2982)					-0.0093 (0.1766)	0.0811 (0.7755)
Advanced Math		0.6485*** (0.2032)					0.6475*** (0.1382)	1.0534* (0.5829)
Committee Score							0.0265 (0.0400)	-0.6809*** (0.1999)
Missing Committee Score							0.2380 (0.1861)	
Year Transition/Matriculate More	0.0710 (0.1280)	0.1591 (0.2057)	-1.1147*** (0.2799)	0.0529 (0.0296, 0.0946)	0.5592** (0.2553)	64.41% (54.59%, 73.15%)	-0.0010 (0.1732)	3.0532*** (0.5423)
Program Rank							-0.0309*** (0.0019)	
$\sqrt{MSE}$		1.9457*** (0.0672)						
Sigma								0.1457*** (0.0234)
Obs Param LL AIC	1000	37	-2755.2	5584.5				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

A5: Simulated Restricted Mat-Dynamic Estimates (Suboptimal)

Table 25b: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-2.0816*** (0.7582)	5.5356*** (1.2750)	-1.4144*** (0.2209)	0.2431 (0.1576, 0.3748)	0.1778 (0.2485)	54.43% (42.33%, 66.04%)	-9.6441*** (1.3938)	-6.4840* (3.3823)	
GRE Quantitative Percentile	0.0230** (0.0090)	-0.0050 (0.0145)					0.1098*** (0.0150)	0.0774** (0.0363)	
Gender	-0.2276* (0.1310)	-0.1115 (0.2034)					0.1637 (0.1438)	0.3953 (0.4595)	
Demographic 1	0.3530** (0.1729)	-0.4625* (0.2685)					-0.4614** (0.2103)	0.0566 (0.6572)	
Demographic 2	0.0922 (0.1908)	-0.2781 (0.2722)					-0.0923 (0.2189)	-0.4297 (0.7030)	
Advanced Math		0.7079*** (0.1960)					0.5798*** (0.1558)	0.7404 (0.5503)	
Committee Score							0.1840*** (0.0433)	-0.5475*** (0.1601)	
Missing Committee Score							0.2500 (0.1936)		
Year Transition/Matriculate More	0.0723 (0.1280)	0.0357 (0.2000)	-1.3730*** (0.3748)	0.0616 (0.0274, 0.1385)	0.5357 (0.3413)	67.12% (53.66%, 78.25%)	-0.0723 (0.1795)	3.1502*** (0.5830)	
Program Rank							-0.0305*** (0.0019)		
$\sqrt{\text{MSE}}$		1.9147*** (0.0677)							
Sigma									0.2346*** (0.0484)
Obs Param LL AIC	1000	37	-2818.0	5710.0					

Robust standard errors in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Confidence intervals are 95%.

A5: Simulated Unrestricted Wait-Dynamic Estimates (Optimal)

Table 26a: Simulated Unrestricted Waitlist-Dynamic Estimates (Optimal)

	Advanced Math		Committee Score		$\bar{m}_{2A}$		Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027** (0.7564)	-0.4663** (0.1830)	4.8637*** (1.3238)	-0.0249 (0.0516)	-11.5885*** (1.0232)	-2.2558*** (0.1454)	-11.9709*** (2.5515)	-1.6176*** (0.3154)	-13.2754*** (2.4119)	-1.6998*** (0.2845)	-9.4807*** (1.3435)	-1.3248*** (0.1562)	-5.9143 (6.4966)	-0.6836 (0.7703)		
GRE Quantative Percentile	0.0208** (0.0090)	0.0051** (0.0022)	0.0008 (0.0150)		0.1236*** (0.0118)	0.0241*** (0.0018)	0.1097*** (0.0271)	0.0148*** (0.0034)	0.1387*** (0.0265)	0.0178*** (0.0032)	0.1150*** (0.0145)	0.0161*** (0.0016)	0.0842 (0.0645)	0.0097 (0.0075)		
Gender	-0.2335* (0.1309)	-0.0572* (0.0319)	-0.0783 (0.2113)		0.2266 (0.1464)	0.0441 (0.0283)	0.0234 (0.2899)	0.0032 (0.0392)	0.3040 (0.3288)	0.0389 (0.0421)	0.2078 (0.1766)	0.0290 (0.0246)	0.4647 (0.9007)	0.0537 (0.1052)		
Demographic 1	0.3691** (0.1732)	0.0905** (0.0421)	-0.3430 (0.2885)		-0.3014 (0.1988)	-0.0587 (0.0386)	0.5534 (0.4942)	0.0748 (0.0665)	-0.8832* (0.4623)	-0.1131* (0.0578)	-0.3380 (0.2456)	-0.0472 (0.0342)	-0.2627 (0.7742)	-0.0304 (0.0879)		
Demographic 2	0.0925 (0.1911)	0.0227 (0.0468)	-0.1806 (0.3001)		0.1538 (0.2215)	0.0299 (0.0431)	0.5137 (0.5205)	0.0694 (0.0700)	-0.0523 (0.4528)	-0.0067 (0.0580)	-0.1354 (0.2652)	-0.0189 (0.0371)	-0.3955 (1.2095)	-0.0457 (0.1391)		
Advanced Math			0.6870*** (0.2024)				0.5067* (0.2907)	0.0685* (0.0390)	0.8622*** (0.3306)	0.1104*** (0.0421)	0.6827*** (0.1770)	0.0954*** (0.0237)	1.1444 (0.9852)	0.1323 (0.1080)		
Committee Score							0.1061 (0.0689)	0.0143 (0.0093)	0.0378 (0.0672)	0.0048 (0.0086)	0.0467 (0.0672)	0.0065 (0.0094)	-0.9116*** (0.3783)	-0.1054*** (0.0317)		
Missing Committee Score											0.2177 (0.1896)	0.0304 (0.0264)				
Year Transition/Matriculate More	0.0664 (0.1281)	0.0163 (0.0314)	0.0932 (0.2069)		0.9840*** (0.1472)	0.1915*** (0.0263)					-0.0466 (0.1740)	-0.0065 (0.0243)	3.3941*** (0.7689)	0.3923*** (0.0698)		
Program Rank											-0.0309*** (0.0019)	-0.0043*** (0.0002)				
$\bar{m}_2$					1.0896*** (0.0851)				-1.5992** (0.7875)	-0.2161** (0.1049)						
$\bar{m}_{2A}$											-2.0178*** (0.7700)	-0.2584*** (0.0964)				
$\sqrt{\text{MSE}}$			1.9465*** (0.0673)	0.0015*** (0.0005)												
Obs Param LL AIC	1000	49	-841.4	1780.9												

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Unrestricted Wait-Dynamic Estimates (Suboptimal)

Table 26b: Simulated Unrestricted Waitlist-Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score		$\overline{m}_{2A}$		Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient		Coefficient		Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027** (0.7564)	-0.4663** (0.1830)	5.2717*** (1.2701)		-0.0496* (0.0259)		-11.2295*** (1.0144)	-2.1801*** (0.1474)	-6.7113*** (2.0007)	-1.0637*** (0.3043)	-7.1639*** (2.5070)	-0.8002*** (0.2796)	-9.7587*** (1.3345)	-1.3686*** (0.1538)	-1.5054 (3.8095)	-0.2179 (0.5534)
GRE Quantative Percentile	0.0208** (0.0090)	0.0051** (0.0022)	-0.0016 (0.0144)				0.1226*** (0.0117)	0.0238*** (0.0018)	0.0387* (0.0215)	0.0061* (0.0034)	0.0717*** (0.0278)	0.0080** (0.0031)	0.1175*** (0.0145)	0.0165*** (0.0016)	0.0362 (0.0435)	0.0052 (0.0063)
Gender	-0.2335* (0.1309)	-0.0572* (0.0319)	-0.1023 (0.2051)				0.2575* (0.1469)	0.0500* (0.0283)	-0.2767 (0.2746)	-0.0439 (0.0431)	0.4102 (0.3417)	0.0458 (0.0380)	0.1984 (0.1766)	0.0278 (0.0247)	-0.3519 (0.5435)	-0.0509 (0.0788)
Demographic 1	0.3691** (0.1732)	0.0905** (0.0421)	-0.4893* (0.2701)				-0.8293*** (0.1966)	-0.1610*** (0.0371)	0.2577 (0.3774)	0.0408 (0.0594)	-1.1548** (0.5110)	-0.1290** (0.0563)	-0.3897 (0.2462)	-0.0547 (0.0343)	0.3291 (0.7092)	0.0476 (0.1031)
Demographic 2	0.0925 (0.1911)	0.0227 (0.0468)	-0.2744 (0.2740)				0.0797 (0.2135)	0.0155 (0.0414)	0.5369 (0.3886)	0.0851 (0.0605)	-0.3696 (0.4652)	-0.0413 (0.0521)	-0.2068 (0.2636)	-0.0290 (0.0370)	-0.7626 (0.7932)	-0.1104 (0.1090)
Advanced Math			0.7219*** (0.1966)						0.2363 (0.2639)	0.0375 (0.0417)	0.4866 (0.3566)	0.0544 (0.0399)	0.6387*** (0.1763)	0.0896*** (0.0238)	0.3977 (0.6444)	0.0576 (0.0922)
Committee Score									0.3737*** (0.0674)	0.0592*** (0.0096)	0.0653 (0.0837)	0.0073 (0.0093)	0.0626 (0.0657)	0.0088 (0.0092)	-0.6714*** (0.1978)	-0.0972*** (0.0217)
Missing Committee Score													0.2535 (0.1896)	0.0356 (0.0264)		
Year Transition/Matriculate More	0.0664 (0.1281)	0.0163 (0.0314)	0.0031 (0.2010)				1.1108*** (0.1479)	0.2156*** (0.0257)					-0.0473 (0.1757)	-0.0066 (0.0247)	2.9420*** (0.6724)	0.4259*** (0.0579)
Program Rank													-0.0303*** (0.0019)	-0.0043*** (0.0002)		
$\hat{m}_2$					0.8882*** (0.0488)				-0.5496 (0.5706)	-0.0871 (0.0900)						
$\overline{m}_{2A}$											-2.9219*** (0.9936)	-0.3264*** (0.1102)				
$\sqrt{MSE}$			1.9186*** (0.0682)		0.0010*** (0.0002)											
Obs Param LL AIC	1000	49	-673.8		1445.7											

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Restricted Wait-Dynamic Estimates (Optimal)

Table 27a: Simulated Restricted Waitlist-Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	$\overline{m}_{2A}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-2.1640*** (0.7576)	4.6955*** (1.3257)	-0.0249 (0.0516)	-0.1957*** (0.0691)	0.8222 (0.7181, 0.9414)	0.2726 (0.2600)	61.72% (51.94%, 70.64%)	-9.4804*** (1.2650)	-5.9143 (6.4966)		
GRE Quantative Percentile	0.0240*** (0.0090)	0.0028 (0.0150)						0.1145*** (0.0138)	0.0842 (0.0645)		
Gender	-0.2130 (0.1303)	-0.0649 (0.2112)						0.2345** (0.1083)	0.4647 (0.9007)		
Demographic 1	0.3565** (0.1724)	-0.3527 (0.2903)						-0.3149** (0.1551)	-0.2627 (0.7742)		
Demographic 2	0.0948 (0.1904)	-0.1805 (0.3010)						0.0030 (0.1591)	-0.3955 (1.2095)		
Advanced Math		0.6919*** (0.2026)						0.6133*** (0.1367)	1.1444 (0.9852)		
Committee Score								0.0498 (0.0381)	-0.9117** (0.3783)		
Missing Committee Score								0.2156 (0.1819)			
Year Transition/Matriculate More	0.0741 (0.1277)	0.0966 (0.2069)		-0.0416 (0.0595)	0.7887 (0.6844, 0.9090)		74.61% (62.50%, 83.82%)	-0.0112 (0.1718)	3.3941*** (0.7689)		
Program Rank								-0.0310*** (0.0019)			
$\overline{m}_2$			1.0896*** (0.0851)								
$\overline{m}_{2A}$						1.2654** (0.6248)					
$\sqrt{\text{MSE}}$		1.9468*** (0.0673)	0.0015*** (0.0005)								
Beta										0.9752*** (0.0346)	
Sigma											0.1383*** (0.0190)
Obs Param LL AIC	1000	33	-873.0	1812.0							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

A5: Simulated Restricted Wait-Dynamic Estimates (Suboptimal)

Table 27b: Simulated Restricted Waitlist-Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	$\overline{m}_{2A}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-2.1752*** (0.7588)	4.5382*** (1.2598)	-0.0496* (0.0259)	-0.1451* (0.0873)	0.8650 (0.7289, 1.0264)	0.5564 (0.3606)	69.21% (54.49%, 80.84%)	-9.3315*** (1.4094)	-1.5054 (3.8095)		
GRE Quantative Percentile	0.0242*** (0.0090)	0.0073 (0.0144)						0.1056*** (0.0150)	0.0362 (0.0435)		
Gender	-0.2183* (0.1306)	-0.0653 (0.2030)						0.2008* (0.1211)	-0.3519 (0.5435)		
Demographic 1	0.3425** (0.1728)	-0.5631** (0.2717)						-0.4888*** (0.1800)	0.3291 (0.7092)		
Demographic 2	0.1007 (0.1906)	-0.2649 (0.2739)						-0.0146 (0.1849)	-0.7626 (0.7932)		
Advanced Math		0.7294*** (0.1970)						0.5146*** (0.1481)	0.3977 (0.6444)		
Committee Score								0.1934*** (0.0379)	-0.6714*** (0.1978)		
Missing Committee Score								0.2645 (0.1891)			
Year Transition/Matriculate More	0.0699 (0.1278)	0.0088 (0.2007)		-0.1153* (0.0612)	0.7708 (0.6517, 0.9116)		83.06% (60.34%, 94.05%)	-0.0422 (0.1751)	2.9420*** (0.6724)		
Program Rank								-0.0305*** (0.0019)			
$\overline{m}_2$			0.8882*** (0.0488)								
$\overline{m}_{2A}$						1.8260 (1.1779)					
$\sqrt{\text{MSE}}$		1.9197*** (0.0684)	0.0010*** (0.0002)								
Beta										0.8510*** (0.0461)	
Sigma											0.1784*** (0.0318)
Obs Param LL AIC	1000	33	-731.0	1528.1							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

A5: Simulated Unrestricted Wait-Hybrid Estimates (Optimal)

Table 28a: Simulated Unrestricted Waitlist-Hybrid Estimates (Optimal)

	$\bar{m}_{2A}$ Coefficient	Transition Coefficient	Transition AME	Accept 2 Coefficient	Accept 2 AME	Accept 3 Coefficient	Accept 3 AME	Success 1 Coefficient	Success 1 AME	Success 3 Coefficient	Success 3 AME	Matriculate Coefficient	Matriculate AME
Intercept	-0.0249 (0.0516)	-11.5885*** (1.0232)	-2.2558*** (0.1454)	-11.9709*** (2.5515)	-1.6176*** (0.3154)	-13.2754*** (2.4119)	-1.6998*** (0.2845)	-8.1374*** (1.1381)	-1.1678*** (0.1344)	-9.4807*** (1.3435)	-1.3248*** (0.1562)	-5.9143 (6.4966)	-0.6836 (0.7703)
GRE Quantitative Percentile		0.1236*** (0.0118)	0.0241*** (0.0018)	0.1097*** (0.0271)	0.0148*** (0.0034)	0.1387*** (0.0265)	0.0178*** (0.0032)	0.1079*** (0.0133)	0.0155*** (0.0015)	0.1150*** (0.0145)	0.0161*** (0.0016)	0.0842 (0.0645)	0.0097 (0.0075)
Gender		0.2266 (0.1464)	0.0441 (0.0283)	0.0234 (0.2899)	0.0032 (0.0392)	0.3040 (0.3288)	0.0389 (0.0421)	0.1399 (0.1739)	0.0201 (0.0249)	0.2078 (0.1766)	0.0290 (0.0246)	0.4647 (0.9007)	0.0537 (0.1052)
Demographic 1		-0.3014 (0.1988)	-0.0587 (0.0386)	0.5534 (0.4942)	0.0748 (0.0665)	-0.8832* (0.4623)	-0.1131* (0.0578)	-0.2533 (0.2359)	-0.0364 (0.0338)	-0.3380 (0.2456)	-0.0472 (0.0342)	-0.2627 (0.7742)	-0.0304 (0.0879)
Demographic 2		0.1538 (0.2215)	0.0299 (0.0431)	0.5137 (0.5205)	0.0694 (0.0700)	-0.0523 (0.4528)	-0.0067 (0.0580)	-0.1179 (0.2581)	-0.0169 (0.0371)	-0.1354 (0.2652)	-0.0189 (0.0371)	-0.3955 (1.2095)	-0.0457 (0.1391)
Advanced Math				0.5067* (0.2907)	0.0685* (0.0390)	0.8622*** (0.3306)	0.1104*** (0.0421)			0.6827*** (0.1770)	0.0954*** (0.0237)	1.1444 (0.9852)	0.1323 (0.1080)
Committee Score				0.1061 (0.0689)	0.0143 (0.0093)	0.0378 (0.0672)	0.0048 (0.0086)			0.0467 (0.0672)	0.0065 (0.0094)	-0.9116** (0.3783)	-0.1054*** (0.0317)
Missing Committee Score										0.2177 (0.1896)	0.0304 (0.0264)		
Year Transition/Matriculate More		0.9840*** (0.1472)	0.1915*** (0.0263)					-0.0587 (0.1690)	-0.0084 (0.0242)	-0.0466 (0.1740)	-0.0065 (0.0243)	3.3941*** (0.7689)	0.3923*** (0.0698)
Program Rank								-0.0310*** (0.0019)	-0.0045*** (0.0002)	-0.0309*** (0.0019)	-0.0043*** (0.0002)		
$\bar{m}_2$	1.0896*** (0.0851)			-1.5992** (0.7875)	-0.2161** (0.1049)								
$\bar{m}_{2A}$						-2.0178*** (0.7700)	-0.2584*** (0.0964)						
$\sqrt{MSE}$	0.0015*** (0.0005)												
Obs Param LL AIC	1000	42	191.4	-298.8									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Unrestricted Wait-Hybrid Estimates (Suboptimal)

Table 28b: Simulated Unrestricted Waitlist-Hybrid Estimates (Suboptimal)

	$\bar{m}_{2A}$ Coefficient	Transition Coefficient	AME	Accept 2 Coefficient	AME	Accept 3 Coefficient	AME	Success 1 Coefficient	AME	Success 3 Coefficient	AME	Matriculate Coefficient	AME
Intercept	-0.0496* (0.0259)	-11.2295*** (1.0144)	-2.1801*** (0.1474)	-6.7113*** (2.0007)	-1.0637*** (0.3043)	-7.1639*** (2.5070)	-0.8002*** (0.2796)	-8.2825*** (1.1392)	-1.1913*** (0.1341)	-9.7587*** (1.3345)	-1.3686*** (0.1538)	-1.5054 (3.8095)	-0.2179 (0.5534)
GRE Quantitative Percentile		0.1226*** (0.0117)	0.0238*** (0.0018)	0.0387* (0.0215)	0.0061* (0.0034)	0.0717*** (0.0278)	0.0080** (0.0031)	0.1097*** (0.0133)	0.0158*** (0.0015)	0.1175*** (0.0145)	0.0165*** (0.0016)	0.0362 (0.0435)	0.0052 (0.0063)
Gender		0.2575* (0.1469)	0.0500* (0.0283)	-0.2767 (0.2746)	-0.0439 (0.0431)	0.4102 (0.3417)	0.0458 (0.0380)	0.1355 (0.1741)	0.0195 (0.0250)	0.1984 (0.1766)	0.0278 (0.0247)	-0.3519 (0.5435)	-0.0509 (0.0788)
Demographic 1		-0.8293*** (0.1966)	-0.1610*** (0.0371)	0.2577 (0.3774)	0.0408 (0.0594)	-1.1548** (0.5110)	-0.1290** (0.0563)	-0.2776 (0.2344)	-0.0399 (0.0337)	-0.3897 (0.2462)	-0.0547 (0.0343)	0.3291 (0.7092)	0.0476 (0.1031)
Demographic 2		0.0797 (0.2135)	0.0155 (0.0414)	0.5369 (0.3886)	0.0851 (0.0605)	-0.3696 (0.4652)	-0.0413 (0.0521)	-0.1809 (0.2566)	-0.0260 (0.0369)	-0.2068 (0.2636)	-0.0290 (0.0370)	-0.7626 (0.7932)	-0.1104 (0.1090)
Advanced Math				0.2363 (0.2639)	0.0375 (0.0417)	0.4866 (0.3566)	0.0544 (0.0399)			0.6387*** (0.1763)	0.0896*** (0.0238)	0.3977 (0.6444)	0.0576 (0.0922)
Committee Score				0.3737*** (0.0674)	0.0592*** (0.0096)	0.0653 (0.0837)	0.0073 (0.0093)			0.0626 (0.0657)	0.0088 (0.0092)	-0.6714*** (0.1978)	-0.0972*** (0.0217)
Missing Committee Score										0.2535 (0.1896)	0.0356 (0.0264)		
Year Transition/Matriculate More		1.1108*** (0.1479)	0.2156*** (0.0257)					-0.0767 (0.1688)	-0.0110 (0.0243)	-0.0473 (0.1757)	-0.0066 (0.0247)	2.9420*** (0.6724)	0.4259*** (0.0579)
Program Rank								-0.0304*** (0.0019)	-0.0044*** (0.0002)	-0.0303*** (0.0019)	-0.0043*** (0.0002)		
$\bar{m}_2$	0.8882*** (0.0488)			-0.5496 (0.5706)	-0.0871 (0.0900)								
$\bar{m}_{2A}$						-2.9219*** (0.9936)	-0.3264*** (0.1102)						
$\sqrt{MSE}$	0.0010*** (0.0002)												
Obs Param LL AIC	1000	42	374.8	-665.5									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

A5: Simulated Restricted Wait-Hybrid Estimates (Optimal)

Table 29a: Simulated Restricted Waitlist-Hybrid Estimates (Optimal)

	$\overline{m}_{2A}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer Intercept	-0.0249 (0.0516)	0.6925*** (0.1818)	66.65% (58.33%, 74.05%)	0.4942 (0.3513)	70.64% (57.55%, 81.02%)	-8.0628*** (1.1622)	-9.5820*** (1.3341)	-5.9143 (6.4966)		
GRE Quantitative Percentile						0.1062*** (0.0134)	0.1154*** (0.0146)	0.0842 (0.0645)		
Gender						0.1999** (0.1018)	0.1696 (0.1358)	0.4647 (0.9007)		
Demographic 1						-0.2482* (0.1353)	-0.3098 (0.1912)	-0.2627 (0.7742)		
Demographic 2						0.0407 (0.1511)	-0.0318 (0.1962)	-0.3955 (1.2095)		
Advanced Math							0.6616*** (0.1424)	1.1444 (0.9852)		
Committee Score							0.0560 (0.0425)	-0.9117** (0.3783)		
Missing Committee Score							0.2167 (0.1838)			
Year Transition/Matriculate More		-0.8491*** (0.2179)	46.09% (39.64%, 52.68%)		88.08% (65.79%, 96.60%)	-0.0604 (0.1667)	-0.0289 (0.1727)	3.3941*** (0.7689)		
Program Rank						-0.0310*** (0.0019)	-0.0310*** (0.0019)			
$\bar{m}_2$	1.0896*** (0.0851)									
$\overline{m}_{2A}$				2.3665* (1.2823)						
$\sqrt{\text{MSE}}$	0.0015*** (0.0005)									
Beta									0.9490*** (0.0391)	
Sigma										0.1926*** (0.0246)
Obs Param LL AIC	1000	26	186.8	-321.6						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

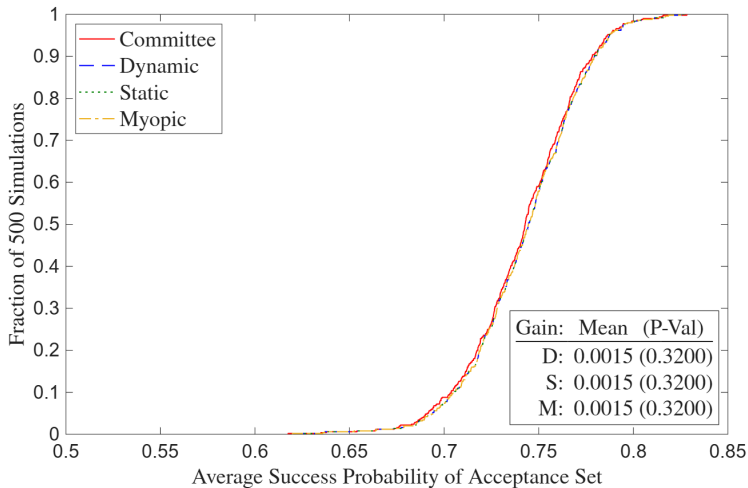
A5: Simulated Restricted Wait-Hybrid Estimates (Suboptimal)

Table 29b: Simulated Restricted Waitlist-Hybrid Estimates (Suboptimal)

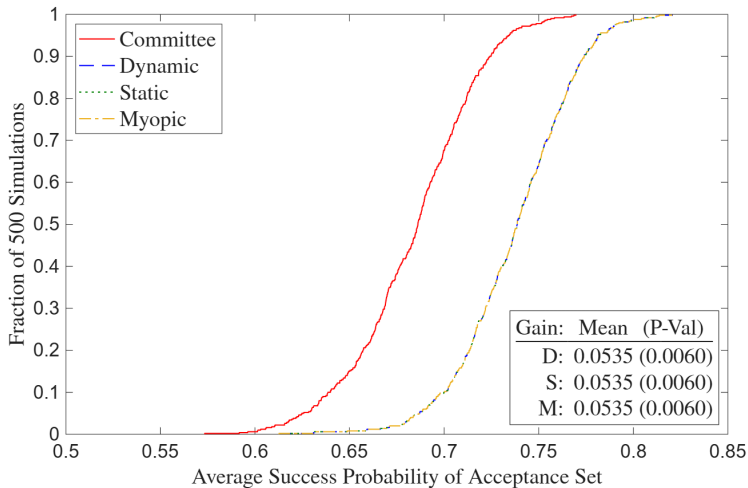
	$\overline{m}_{2A}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer Intercept	-0.0496* (0.0259)	0.8342*** (0.2171)	69.72% (60.07%, 77.90%)	0.8216 (0.7603)	80.71% (61.39%, 91.67%)	-8.6331*** (1.1831)	-8.3628*** (1.4735)	-1.5054 (3.8095)		
GRE Quantitative Percentile						0.1147*** (0.0137)	0.0947*** (0.0156)	0.0362 (0.0435)		
Gender						0.2301** (0.1099)	0.1269 (0.1412)	-0.3519 (0.5435)		
Demographic 1						-0.5688*** (0.1473)	-0.3599* (0.2046)	0.3291 (0.7092)		
Demographic 2						-0.0225 (0.1609)	-0.0641 (0.2095)	-0.7626 (0.7932)		
Advanced Math							0.5402*** (0.1471)	0.3977 (0.6444)		
Committee Score							0.1927*** (0.0410)	-0.6714*** (0.1978)		
Missing Committee Score							0.1774 (0.1951)			
Year Transition/Matriculate More		-1.0924*** (0.2510)	43.58% (37.09%, 50.29%)		96.46% (13.07%, 99.98%)	-0.0704 (0.1661)	-0.0765 (0.1767)	2.9420*** (0.6724)		
Program Rank						-0.0307*** (0.0019)	-0.0305*** (0.0019)			
$\bar{m}_2$	0.8882*** (0.0488)									
$\overline{m}_{2A}$				4.3874 (5.6885)						
$\sqrt{\text{MSE}}$	0.0010*** (0.0002)									
Beta									0.8225*** (0.0434)	
Sigma										0.2234*** (0.0318)
Obs Param LL AIC	1000	26	351.0	-650.0						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

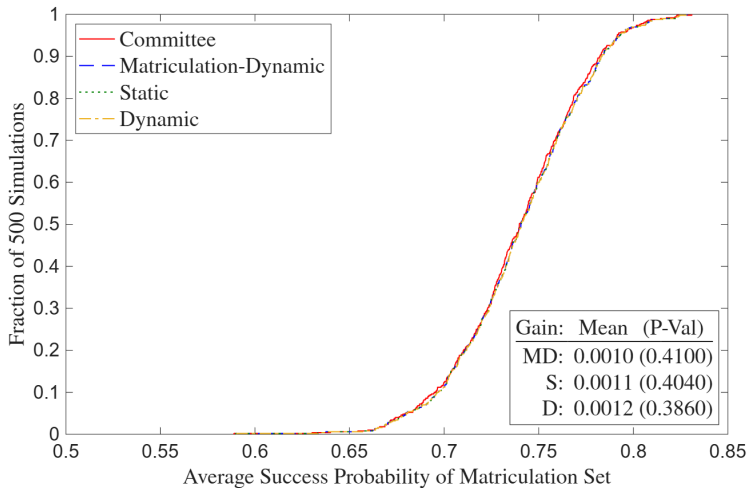
A5: Simulated Deviation Gains (Optimal)



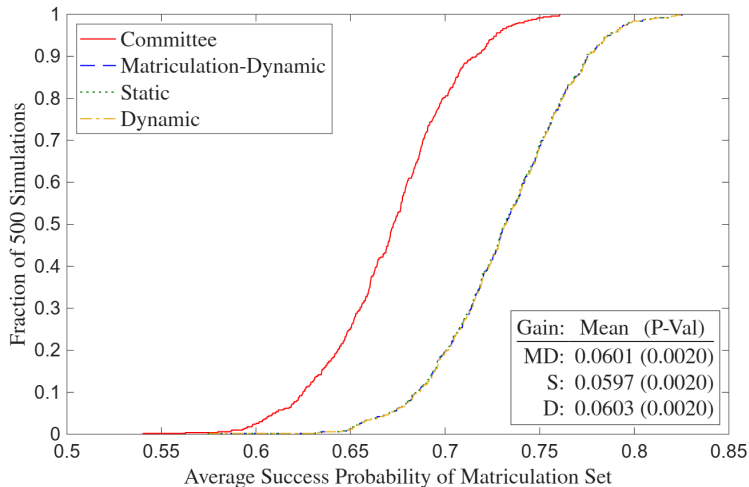
A5: Simulated Deviation Gains (Suboptimal)



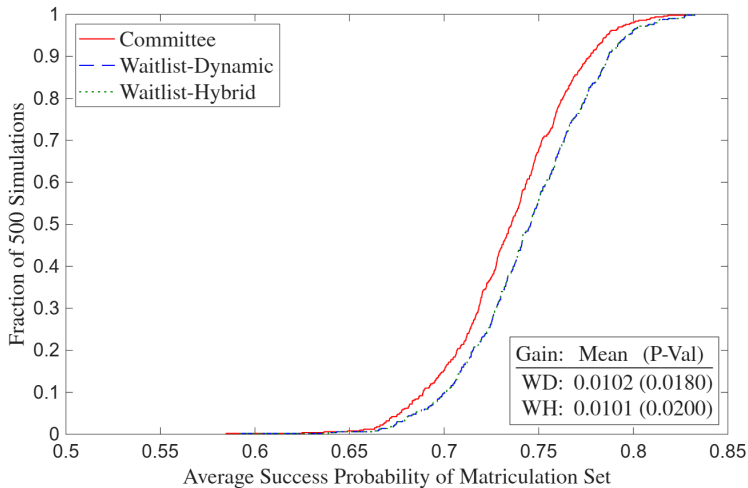
A5: Simulated Matriculation Deviation Gains (Optimal)



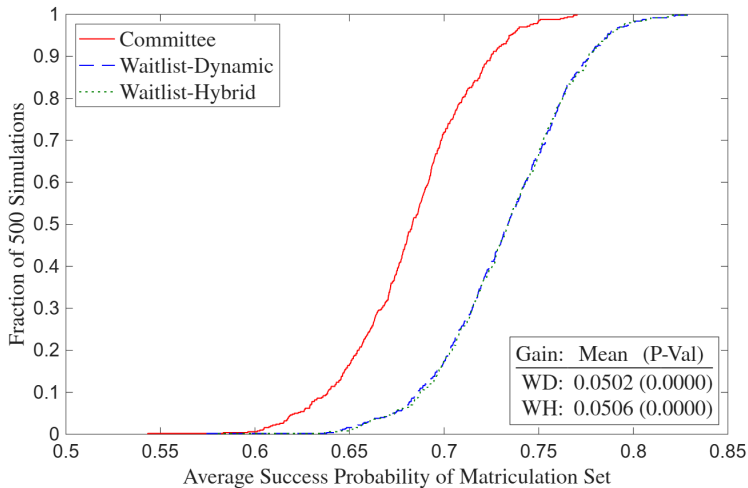
A5: Simulated Matriculation Deviation Gains (Suboptimal)



A5: Simulated Waitlist Deviation Gains (Optimal)



A5: Simulated Waitlist Deviation Gains (Suboptimal)



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